

TESTING OF HYPOTHESIS (LARGE SAMPLE TESTS)

INTRODUCTION

The outcome of a statistical experiment may be recorded either as a numerical value or as a descriptive representation.

When a pair of dice are tossed and the sum of the numbers on the faces is the outcome of interest, we record a numerical value. However, if the students of a certain school are given blood tests and the type of blood is of interest, then a descriptive representation might be most useful. A person's blood can be classified in 8 ways. It must be AB , A , B or O , with a plus or minus sign, depending on the presence or absence of the Rh antigen.

The statistician is primarily concerned with the analysis of numerical data.

For the classification of blood types, it may be convenient to use numbers from 1 to 8 to represent the blood types and then record the appropriate number for each student.

In any particular study, the number of possible observations may be small, large but finite, or infinite. For example, in the classification of blood types we can only have as many observations as there are students in the school. The project, therefore results in a finite number of observations.

On the other hand, if we could toss a pair of dice indefinitely and record the sums that occur, we would obtain an infinite set of values, each value representing the result of a single toss of a pair of dice.

In this chapter we focus on sampling from distributions or populations and study such important quantities as the **sample mean** and **sample variance**.

[JNTU (A) Dec. 2018]

12 POPULATION AND SAMPLE

We begin this section by discussing the notions of population and sample.

The totality of observations with which we are concerned, whether this number be finite or infinite, constitutes what we call *population*. There was a time when the word *population* referred to observations obtained from statistical studies about people. Today, the statistician uses the term to refer to groups of people, animals, or all possible outcomes from some complicated biological or engineering system or numerical data. Thus the population does not imply living beings alone, e.g., we speak of the population of births, heights, weights, prices of vegetables and so on.

Definition : Population (or universe) is the aggregate or totality of statistical data forming a subject of investigation. For example,

- (i) the population of the heights of Indians,
- (ii) the population of Nationalised Banks in India, etc.

Definition : The number of observations in the population is defined to be the *size of the population*. It may be finite or infinite. Size of population is denoted by N .

Examples : If there are 600 students in the school that we classified according to blood type, we say that we have a population of size 600. The numbers on the cards in a deck, the heights of residents in a certain city, and the lengths of fish in a particular lake are all examples of population with *finite* size.

The observations obtained by measuring the atmospheric pressure every day from the past on into the future, or all measurements on the depth of a lake from any conceivable position, are examples of population whose sizes are infinite.

In the field of statistical inference the statistician is interested in arriving at conclusions concerning a population when it is impossible or impractical to observe the entire set of observations that make up the population.

For example, in attempting to determine the average length of life of a certain brand of light bulb, it would be impossible to test all such bulbs if we are to have any left to sell.

Therefore, we must depend on a subset of observations from the population to help us make inferences concerning the same population. This brings us to consider the notion of sampling.

Sampling : Most of the times, study of entire population may not be possible to carry out and hence a part alone is selected from the given population. A portion of the population which is examined with a view to determining the population characteristics is called a sample. *i.e.*, a sample is a subset of population and the number of objects in the sample is called the size of the sample. Size of the sample is denoted by n .

The process of selection of a sample is called sampling. It is quite often used in our day-to-day practical life. For example, (i) to assess the quality of a bag of rice, sugar, wheat or any other commodity, we examine only a portion of it by taking a handful of it from the bag and then decide to purchase it or not. The portion selected from the bag is called a sample, while the whole quantity of rice, sugar or wheat in the bag is the population, (ii) to estimate the proportion of defective articles in a large consignment, only a portion (*i.e.*, a few of them) is selected and examined. The portion selected is a sample (iii) cars produced in India is the population and the Nano cars is the sample.

When information is collected in respect of every individual item, the enquiry is said to be done by *Complete Enumeration* or *Census*. For example, during the *Census of Population* (which is done every ten years in India), information in respect of each individual person residing in India is collected. This method gives information for each and every unit of the population with greater accuracy. But this method involves multiplicity of causes viz., administrative and financial implications, time factor, etc. In most cases of statistical inquiry, because of limitations of time and cost, only a portion (*i.e.*, a sample) of the available source of information is examined, and the data collected from them. This process of partial enumeration is known as *Sample Survey*. The results are then generalised and made applicable to the whole field of inquiry. This is known as *Sampling*.

Thus in estimating the characteristics of the population, instead of enumerating entire population, only the individuals in the sample are examined. Then the sample characteristics are utilised to approximately estimate the population. The error involved in such approximation is known as *sampling error* and is inherent and unavoidable in any and

every sampling scheme. However the advantage of sampling process lies in considerable gains in time, cost and handling of large data.

The population is thus a universal set for the sample. The statistical constants like mean, standard deviation, correlation coefficient, etc. obtained for the population are called parameters.

Similarly, constants for the sample drawn from the given population *i.e.*, mean ($\bar{x}$), standard deviation (S) etc., are called the statistic.

If our inferences from the sample to the population are to be valid, we must obtain samples that are representative of the population. All too often, we are tempted to choose a sample by selecting the most convenient members of the population. Such a procedure may lead to erroneous inferences concerning the population. Any sampling procedure that produces inferences that consistently over estimate (or) consistently under estimate some characteristics of the population is said to be biased. To eliminate any possibility of bias in the sampling procedure it is desirable to choose a random sample in the sense that the observations are made independently and at random.

9.3 DIFFERENT METHODS OF SAMPLING

Some important methods of sampling are discussed below.

I. Probability Sampling Methods

1. Random Sampling (or Probability Sampling)

It is the process of drawing a sample from a population in such a way that each member of the population has an equal chance of being included in the sample. The sample obtained by the process of random sampling is called a random sample.

For example : (i) A handy of cards dealt from a well - shuffled pack of cards is a random sample.

(ii) Selecting randomly 20 words from a dictionary is a random sample.

(iii) Choosing 10 patients from a hospital in order to test the efficacy of a certain newly - invented drug.

If each element of a population may be selected more than once then it is called **sampling with replacement** whereas if the element cannot be selected more than once, it is called **sampling without replacement**.

Note : If N is the size of a population and n is the sample size, then

(i) The number of samples with replacement = N^n

(ii) The number of samples without replacement = ${}^N C_n$

2. Stratified Sampling (or Stratified Random Sampling)

This method is useful when the population is heterogeneous. In this type of sampling, the population is first sub-divided into several parts (or small groups) called *strata* according to some relevant characteristics so that each stratum is more or less homogeneous. Each stratum is called a sub-population. Then a small sample (called sub-sample) is selected from each stratum at random. All the sub-samples are combined together to form the stratified sample which represents the population properly. The process of obtaining and examining a stratified sample with a view to estimating the characteristic of the population is known as *Stratified Sampling*.

For example, let us select a stratified sample of 500 families from a city having 50,000 families, with a view to studying their economic condition. For this purpose, the city area is divided into a number of strata, according to economic condition of their inhabitants, as measured by annual income (say). Thus, localities mostly inhabited by people with more or less similar annual income may be included under one stratum. A few families are then chosen at random from each so that the sum total of all the families from all the strata is 500.

3. Systematic Sampling (or Quasi - Random Sampling)

As the name suggests this means forming the sample in some systematic manner by taking items at regular intervals. In this method, all the units of the population are arranged in some order. If the population size is finite, all the units of the population are arranged in some order. Then from the first k items, one unit is selected at random. This unit and every k th unit of the serially listed population combined together constitute a systematic sample. This type of sampling is known as *Systematic Sampling*.

The difference between random sampling and systematic sampling lies in the fact that in the case of a random sample all the members have to be chosen randomly, whereas in the case of a systematic sample only the first member has to be chosen at random.

II. Non-Probability Sampling Methods.

4. Purposive Sampling (or Judgement Sampling)

When the choice of the individual items of a sample entirely depends on the individual judgement of the investigator (or sampler), it is called a *Purposive or Judgement Sampling*.

In this method, the members constituting the sample are chosen not according to some definite scientific procedure, but according to convenience and personal choice of the individual, who selects the sample. Two or more such independent purposive samples may give widely different estimates of the same population. In this type, the investigator must have a good deal of experience and a thorough knowledge of the population. Purposive selection is always subject to some kind of bias. This method is suitable when the sample is small.

For example, if a sample of 20 students is to be selected from a class of 100 to analyse the extra-curricular activities of the students, the investigator would select 20 students who, in his judgement, would represent the class.

5. Sequential Sampling

It consists of a sequence of sample drawn one after another from the population depending on the results of previous samples. If the result of the first sample leads to a decision which is not acceptable, the lot from which the sample was drawn is rejected. But if the result of the first sample is acceptable, no new sample is drawn. But if the first sample leads to no clear decision, a second sample is drawn and, as before, if required, a third sample is drawn to arrive at a final decision to accept or reject the lot. It is widely used in Statistical Quality Control in factories engaged in mass production and other areas.

9.4 CLASSIFICATION OF SAMPLES

[JNTU (A) June 2018]

Samples are classified in two ways.

1. **Large sample** : If the size of the sample (n) ≥ 30 , the sample is said to be large sample.

2. **Small sample** : If the size of the sample (n) < 30 , the sample is said to be small sample (or) exact sample.

In sampling *with replacement*, each member of the population may be chosen more than once, since the member is replaced in the population.

Thus sampling from finite population *with replacement* can be considered theoretically as sampling from *infinite population*. Whereas, in sampling *without replacement*, an element of the population cannot be chosen more than once, as it is not replaced. Therefore the sampling from finite population *without replacement* can be considered theoretically as sampling from finite population only.

Definition : Let $X_1, X_2, \dots, X_n$ be n independent random variables each having the same probability distribution $f(x)$. We then define $X_1, X_2, \dots, X_n$ to be a random sample of size n from the population $f(x)$ and write its joint probability distribution as

$$f(x_1, x_2, \dots, x_n) = f(x_1) \cdot f(x_2) \cdot \dots \cdot f(x_n)$$

Our main purpose in selecting random sample is to elicit information about the unknown population parameters.

Statistic : It is a real valued function of the random sample. So statistic is a function of one or more random variables not involving any unknown parameter. Therefore statistic is a random variable and has a probability or frequency distribution.

9.5 PARAMETERS AND STATISTICS

Parameter is a statistical measure based on all the units (or observations) of a population. Statistic (or sample statistic) is a statistical measure based only on all the units selected in a sample. For example, the statistical constants of the population namely mean μ , variance σ^2 are usually referred to as *parameters*, statistical measures computed from the sample observations alone e.g., sample mean ($\bar{x}$), sample variance (s^2) etc., are usually referred to as *statistics*.

In other words, the mean, median, mode, standard deviation, variance measures of the population are called **parameters** and the measures obtained from the sample of the population are called **statistics**.

Thus *parameters* refer to population while *statistics* refer to sample.

Note : As the units selected in two or more samples drawn from a population are not the same, the value of a statistic varies from sample to sample, but the parameter always remains constant (since all the units in a population remain the same). This variation in the value of a statistic is called *sampling fluctuation*. A parameter has no sampling fluctuation. Usually, statistic is used to estimate the value of an unknown parameter obtained from the sample, is a function of the sample values only.

1. The Sample Mean :

Definition : If $X_1, X_2, \dots, X_n$ represent a random sample of size n , then the sample mean is defined by the statistic

$$\bar{X} = \sum_{i=1}^n \frac{X_i}{n}$$

Note that the statistic $\bar{x}$ has the value $\bar{x} = \sum_{i=1}^n \frac{x_i}{n}$ when x assumes the values $x_1, x_2, \dots$. For instance, the term sample mean is applied to both the statistic $\bar{X}$ and its computed value $\bar{x}$.

2. The Sample Variance :

Definition : If $X_1, X_2, \dots, X_n$ represent a random of size n , then the sample variance is defined by the statistic $s^2 = \sum_{i=1}^n \frac{(X_i - \bar{X})^2}{n-1} = \frac{n \sum X_i^2 - (\sum X_i)^2}{n(n-1)}$

(measure of variability of data about the mean)

Note that s^2 is essentially defined to be the average of the squares of the deviations of the observations from their mean. With the change that the sum of the squared deviations is divided by $n-1$ and not n .

Ex 1 : If a sample size 5 results in the sample values of 8, 5, 9, 6 and 2, find the sample mean. [JNTU (A) June 2019]

Solution : We know that mean $= \bar{x} = \frac{\sum_{i=1}^n x_i}{n}$. Here $n = 5$.

$$\therefore \text{Sample mean} = \frac{x_1 + x_2 + x_3 + x_4 + x_5}{5} = \frac{8 + 5 + 9 + 6 + 2}{5} = \frac{30}{5} = 6$$

Ex 2 : A comparison of coffee prices at 4 randomly selected grocery stores in a city showed increases from the previous month of 12, 15, 17 and 20 rupees for 1 kg. Find the variance of this sample of price increases.

Solution : Calculating the sample mean, we get

$$\bar{x} = \frac{12 + 15 + 17 + 20}{4} = \text{Rs. } 16/-$$

$$\begin{aligned} \therefore s^2 &= \sum_{i=1}^4 \frac{(x_i - 16)^2}{4-1} \\ &= \frac{(12-16)^2 + (15-16)^2 + (17-16)^2 + (20-16)^2}{3} \\ &= \frac{(-4)^2 + (-1)^2 + (1)^2 + (4)^2}{3} = \frac{34}{3} \end{aligned}$$

3. The Sample Standard Deviation :

The sample S. D., denoted by s , is the positive square root of the sample variance.

Note : In the above example 2 S.D., $s = \sqrt{34/3}$.

9.6 SAMPLING DISTRIBUTION

Sampling theory is the study of relationships between a population and samples drawn from the population, and it is applicable to random samples only. We need to obtain maximum information about the population with the help of samples, i.e., to determine the true value of the population parameters (population mean, population S. D., population proportion, etc.) by using sample statistics like sample mean, sample S. D., sample

proportion, etc., and to find the limits of accuracy of estimates based on samples. It helps us to determine whether the differences between two samples are actually due to chance variation or whether they are really significant. Sampling theory also useful in testing of hypothesis and significance which is important in the theory of decisions.

9.7 SAMPLING DISTRIBUTION OF A STATISTIC

Sampling distribution of a statistic is the frequency distribution which is formed with various values of a statistic computed from different samples of the same size drawn from the same population. We can draw a large number of samples of same size from a population of fixed size, each sample containing different population members. Any statistical measure of sample (like mean, median, variance, etc.) may be computed for each of these samples. As a result a series of various values of that statistic may be obtained. These various values can be arranged into a frequency distribution table, which is known as the sampling distribution of the statistic.

Sampling may be done with replacement or without replacement. Sampling with replacement means that the same unit of the population may be included in each sample more than once. Sampling without replacement means that the same unit of population may not be included in each sample more than once.

Let us consider a finite population of size N and let us draw all possible random samples each of the same size n . Then we get ${}^N C_n = \frac{N!}{n!(N-n)!} = k$ (say) samples.

Compute a statistic t (such as sample mean, S.D., etc) for each of these samples. The value of a statistic t may vary from sample to sample. Let $t_1, t_2, \dots, t_k$ be the values of statistic t for the k samples. Each of these values occur with a definite probability. Thus we can construct a table showing the set of values $t_1, t_2, \dots, t_k$ of t with their respective probabilities. This probability distribution of t is known as the Sampling Distribution of t . Thus sampling distribution describes how a statistic t will vary from one sample to the other of the same size. Although all the k samples are drawn from the given population, the members included in different samples are different.

If N is large, then the number k of all possible samples is also large (i.e., sampling without replacement). In such a case, the values $t_1, t_2, \dots, t_k$ of t may be arranged in the form of a relative frequency distribution and the limiting form of this relative frequency distribution when $k \rightarrow \infty$ is called the sampling distribution of statistic t . A statistic (i.e., sample mean, sample S.D, etc.) has always a sampling distribution, but a parameter (i.e. population mean, population S.D.) has no sampling distribution.

If the statistic t is mean, then the corresponding distribution of the statistic is known as sampling distribution of means.

The sample mean can be regarded as a random variable $\bar{x}$ and each sample mean then constitute as the observed value of this new random variable $\bar{x}$. Let these values be: $\bar{x}_1, \bar{x}_2, \bar{x}_3, \dots, \bar{x}_n$. These mean values $\bar{x}_1, \bar{x}_2, \bar{x}_3, \dots, \bar{x}_n$ can be used to form a frequency distribution. Then this frequency distribution of the statistic $\bar{x}$ is known as the sampling distribution.

distribution of the sample mean. Similarly the sampling distribution of standard deviation or variance may be constructed with the various values of standard deviation or variance respectively.

The main characteristic of the Sampling distribution of a statistic is that it approaches normal distribution even when the population distribution is not normal provided the sample size is sufficiently large (greater than 30). Another important feature of the sampling distribution of statistic is that the mean and the standard deviation of the sampling distribution of sample mean bear a definite relation to the corresponding parameters i.e., mean and standard deviation of parent population. These characteristics of the sampling distribution help us:

- (i) To estimate the unknown population parameter from the known statistic
- (ii) To set the confidence limits of the parameter within which the parameter values are expected to lie.
- (iii) To test a hypothesis and to draw a statistical inference from it.

9.8 CENTRAL LIMIT THEOREM

If $\bar{x}$ be the mean of a random sample of size n drawn from a population having mean μ and S.D. σ , then the sampling distribution of the sample mean $\bar{x}$ is approximately a normal distribution with mean μ and S.D. = S.E. of $\bar{x} = \frac{\sigma}{\sqrt{n}}$ provided the sample size n is large ($n \geq 30$). This is established by central limit theorem stated below (without proof).

Theorem : If $\bar{x}$ be the mean of a sample size n drawn from a population with mean μ and S. D. σ then the standardized sample mean

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

is a random variable whose distribution function approaches that of the standard normal distribution $N(z; 0, 1)$ as $n \rightarrow \infty$.

9.9 STANDARD ERROR (S. E.) OF A STATISTIC

The standard error of a statistic t (i.e., S. E. of sample mean, or sample S.D.) is the standard deviation of the sampling distribution of the statistic. Thus S.E. of sample mean $\bar{x}$ is the S.D. of the sampling distribution of sample mean. It is used for assessing the difference between the expected value and observed value. It plays an important role in large sample theory and forms the basis in tests of hypothesis or tests of significance. It gives an idea about the reliability and precision of a sample. S. E. enables us to determine the confidence limits within which the parameters are expected to lie. For example, the probable limits for population proportion P are given by $p \pm 3 \cdot \sqrt{pq/n}$.

The standard errors (S.E.) of some of the well-known statistics are given below (without proof), where μ the population mean, σ^2 the population variance, P the population proportion, n is the sample size, $\bar{x}$ the sample mean, s^2 the sample variance and p the sample proportion.

Note : S.E. of a statistic may be reduced by increasing sample size n , but this results in corresponding increase in cost, time and labour, etc.

Formulae for S.E.

1. S.E. of sample mean $\bar{x} = \frac{\sigma}{\sqrt{n}}$. It is written as S. E. ($\bar{x}$) = $\frac{\sigma}{\sqrt{n}}$

2. S.E. of sample proportion $p = \sqrt{\frac{PQ}{n}}$, where $Q = 1 - P$

3. S.E. of sample S. D (s) = $\frac{\sigma}{\sqrt{2n}}$

4. S.E. of the difference of two sample means $\bar{x}_1$ and $\bar{x}_2$

i.e., S.E. of $(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$ where $\bar{x}_1$ and $\bar{x}_2$ are the means of two random samples of sizes n_1 and n_2 drawn from two populations with S.D. σ_1 and σ_2 respectively.

5. S.E. of $(p_1 - p_2) = \sqrt{\frac{P_1Q_1}{n_1} + \frac{P_2Q_2}{n_2}}$, where p_1 and p_2 are the proportions of two random samples of sizes n_1 and n_2 drawn from two populations with proportions P_1 and P_2 respectively.

6. S.E. of $(s_1 - s_2) = \sqrt{\frac{\sigma_1^2}{2n_1} + \frac{\sigma_2^2}{2n_2}}$

For a finite population of size N , when a sample is drawn without replacement, we have

(i) S.E. of sample mean = $\frac{\sigma}{\sqrt{n}} \cdot \sqrt{\frac{N-n}{N-1}}$

(ii) S.E. of sample proportion = $\sqrt{\frac{PQ}{n}} \cdot \sqrt{\frac{N-n}{N-1}}$

For an infinite population when the sample is drawn without replacement, Formulae 1 and 2 remain the same.

Proposition (Statement without proof) : The mean and S. E. of sample mean $\bar{x}$ (i.e., mean and S.D. of the sampling distribution of $\bar{x}$) when samples of the same size n are drawn from a population having mean μ and S.D. σ are given by Mean of $\bar{x} = E(\bar{x}) = \mu$

and S.E. of $\bar{x} = \frac{\sigma}{\sqrt{n}}$.

9.10 SAMPLING DISTRIBUTION OF MEAN (σ KNOWN)

The probability distribution of $\bar{X}$ is called the sampling distribution of means.

The sampling distribution of a statistic depends on the size of the population, the size of the samples, and the method of choosing the samples.

Let $X_1, X_2, X_3, \dots, X_n$ be the n random samples drawn from a population of size N with mean μ and variance σ^2 and $\bar{X}$ is the mean of samples.

Infinite Population : Suppose the samples are drawn from an infinite population i.e., $N \rightarrow \infty$ (or) sampling is done with replacement, then

(i) The mean of the sampling distribution of means, $\mu_{\bar{X}} = \mu$ i.e., $E(\bar{X}) = \mu$

Proof: We have $\bar{X} = \frac{X_1 + X_2 + X_3 + \dots + X_n}{n}$

Since $\bar{X}$ is a linear combination of the random variables $X_1, X_2, X_3, \dots, X_n$, therefore it is also a random variable.

$$\therefore \text{Mean of } \bar{X} = E(\bar{X}) = \frac{1}{n} E(X_1 + X_2 + X_3 + \dots + X_n)$$

$$= \frac{1}{n} [E(X_1) + E(X_2) + E(X_3) + \dots + E(X_n)]$$

$$= \frac{1}{n} [\mu + \mu + \mu + \dots + \mu] = \frac{n\mu}{n} = \mu$$

(ii) Variance, $\sigma_{\bar{X}}^2 = \frac{\sigma^2}{n}$ i.e., $\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$

$$\therefore \text{S.D of mean, } \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

Proof: If the sample is drawn with replacement, then $X_1, X_2, X_3, \dots, X_n$ are independent random variables. Thus we have $\text{Var}(\bar{X}) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$.

The sampling distribution of $\bar{x}$ will be approximately *normal* with mean μ and variance $\frac{\sigma^2}{n}$ provided that the sample size is large.

$$\text{Standardized sample mean, } z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

Finite Population :

Consider a finite population of size N with mean μ and standard deviation σ . Draw all possible samples of size n without replacement, from this population. Then

(i) The mean of the sampling distribution of means (for $N > n$) is $\mu_{\bar{X}} = \mu$.

$$\text{i.e., } E(\bar{X}) = \mu$$

(ii) The variance is $\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n} \cdot \left(\frac{N-n}{N-1} \right)$

i.e., $\text{Var}(\bar{X}) = \frac{N-n}{N-1} \cdot \frac{\sigma^2}{n}$ (if the sample is drawn without replacement)

and S.D is $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \cdot \sqrt{\frac{N-n}{N-1}}$

Here, the factor $\left(\frac{N-n}{N-1} \right)$, often called the finite population correction factor.

We note that this term tends to become closer and closer to unity as population size becomes larger and larger.

Normal Population (Small Sample) :

Sampling distribution of $\bar{X}$ is normally distributed even for small samples of size $n < 30$ provided sampling is from normal population.

Non - Normal Population (Large Sample) :

Consider a population with unknown (non-normal) distribution. Let the population mean μ and population variance σ be both finite. Let the population be finite or infinite. In case the population is finite assume that the population size N is at least twice the sample size n . Draw all possible samples of size n . Then the sampling distribution of $\bar{X}$ is approximately normally distributed with mean $\mu_{\bar{x}} = \mu$ and variance

$\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}$ provided the sample size is large (i.e., $n \geq 30$).

SOLVED EXAMPLES

Example 1 : If $\bar{x} = 31.45, \mu = 32.3, \sigma = 1.6$ and $n = 40$, find z-test statistic.

Solution : We have $z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} \therefore z = \frac{31.45 - 32.3}{1.6 / \sqrt{40}} = -6.32$ [JNTU (A) June 2019]

Example 2 : (i) What is the value of correction factor if $n = 5$ and $N = 200$.

(ii) The size of the population is 2000 and the size of the sample is 200. Find the correction factor in the population. [JNTU (A) June 2019]

Solution : (i) Given $N = \text{Size of the finite population} = 200$
 $n = \text{Size of the sample} = 5$

$\therefore \text{Correction factor} = \frac{N-n}{N-1} = \frac{200-5}{200-1} = \frac{195}{199} = 0.98$

(ii) This is left as an exercise to the reader.

Example 3 : Find the value of the finite population correction factor for $n = 10$ and $N = 100$. [JNTU 1999, 2000S, (A) Dec. 2009, Nov. 2010, Apr. 2012 (Set No. 3)]

Solution : Given $N = \text{Size of the finite population} = 1000$
 $n = \text{Size of the sample} = 10$

$$\therefore \text{Correction factor} = \frac{N-n}{N-1} = \frac{1000-10}{1000-1} = \frac{990}{999} = 0.991$$

Example 4 : How many different samples of size two can be chosen, from a finite population of size 25.

Solution : We can take ${}^N C_n$ samples of size n from the population of size N .

Here $N = 25$, $n = 2$

$\therefore$ We can take ${}^{25} C_2 = 300$ samples of size 2 from finite population of size 25.

9.11 TEST OF HYPOTHESIS

We need to decide whether to accept or reject a statement about the parameter. This statement is called a *hypothesis* and the decision-making procedure about the hypothesis is called *hypothesis testing*. This is one of the most useful aspects of statistical inference, since many types of decision-making problems, tests or experiments in the engineering world can be formulated as hypothesis - testing problems.

The main object of the Sampling Theory is the study of the Tests of Hypothesis or Tests of Significance.

In many circumstances, we are to make decisions about population, on the basis of only sample information. For example, on the basis of sample data,

- (i) a drug chemist is to decide whether a new drug is really effective in curing a disease,
- (ii) a quality control manager is to determine whether a process is working properly,
- (iii) a statistician has to decide whether a given coin is biased, etc.

Such decisions are called statistical decisions (or simply decisions)

Definition : Statistical Hypothesis

In many circumstances, to arrive at decisions about the population on the basis of sample information, we make assumptions (or guesses) about the population parameters involved. Such an assumption (or statement) is called a *Statistical Hypothesis* which may or may not be true. The procedure which enables us to decide on the basis of sample results whether a hypothesis is true or not, is called Test of Hypothesis or Test of Significance.

Since we use probability distributions to represent population, a statistical hypothesis may also be thought of as a statement about the probability distribution of a random variable. The hypothesis will usually involve one or more parameters of this distribution.

Example : 1. The majority of men in the city are smokers.

2. The teaching methods in both the schools are effective.

9.12 PROCEDURE FOR TESTING A HYPOTHESIS [JNTU NOV. 2008 (SET NO. 2)]

Test of Hypothesis involves the following steps :

Step 1 : Statement (or assumption) of Hypothesis

There are two types of hypothesis :

- (i) Null Hypothesis
- (ii) Alternative Hypothesis.

Null Hypothesis : For applying the tests of significance, we first set up a hypothesis - a definite statement about the population parameter. Such a hypothesis is usually a hypothesis of no-difference, is called Null Hypothesis.

It is in the form $H_0 : \mu = \mu_0$

μ_0 is the value which is assumed or claimed for the population characteristic. It is the reference point against which the Alternative Hypothesis is set up, as explained in the next step.

Definition : A null hypothesis is the hypothesis which asserts that there is no significant difference between the statistic and the population parameter and whatever observed difference is there, is merely due to fluctuations in sampling from the same population. It is always denoted by H_0 . To test whether one procedure is better than another, we assume that there is no difference between the procedures. Similarly to test whether there is a relationship between two variates, we take H_0 that there is no relationship.

For example, in case of a single statistic, H_0 will be that the sample statistic does not differ significantly from the hypothetical parameter value and in the case of two statistics (H_0) will be that the sample statistics do not differ significantly.

(ii) Alternative Hypothesis : Any hypothesis which contradicts the Null Hypothesis is called an *Alternative Hypothesis*, usually denoted by H_1 . The two hypothesis H_0 and H_1 are such that if one is true, the other is false and vice versa. For example, if we want to test the null hypothesis that the population has a specified mean μ_0 (say) i.e., $H_0 : \mu = \mu_0$, then the Alternative Hypothesis would be

(i) $H_1 : \mu \neq \mu_0$ (i.e., either $\mu > \mu_0$ or $\mu < \mu_0$)

or (ii) $H_1 : \mu > \mu_0$

or (iii) $H_1 : \mu < \mu_0$

The Alternative Hypothesis (i) is known as a two-tailed alternative and the Alternative Hypothesis in (ii) is known as right-tailed and in (iii) is known as left-tailed.

The setting of alternative hypothesis is very important to decide whether we have to use a single-tailed (right or left) or two-tailed test.

Alternate Hypothesis is in one of the following forms.

$H_1 : \mu \neq \mu_0$

or $H_1 : \mu > \mu_0$

or $H_1 : \mu < \mu_0$

One has to choose from the above three forms depending on the situation posed, as explained below.

In the example relating to the heights of students, the situation involved only testing the statement made about the average height of a class of students. Therefore, the Alternative Hypothesis was of the form $\mu \neq \mu_0$.

In the example about the life of electric bulbs, the company manager wanted to test whether the average life was more than 560 hrs. Therefore the Alternative Hypothesis was of the form $\mu > \mu_0$.

Consider another example, the manufacturer of cigarettes ordered that the nicotine content in a cigarette should not exceed the stipulated level of 25 mgs. In order to check this, he selected a sample of 100 cigarettes from the lot to be packed, and found that the average nicotine content was 23.5 mgs. Could he be reasonably sure that the stipulations laid down by him were being met? In this case, the manufacturer wanted the nicotine content to be less than a particular value, and therefore, the Alternative Hypothesis is of the form $\mu < \mu_0$.

Step 2 : Specification of the Level of Significance

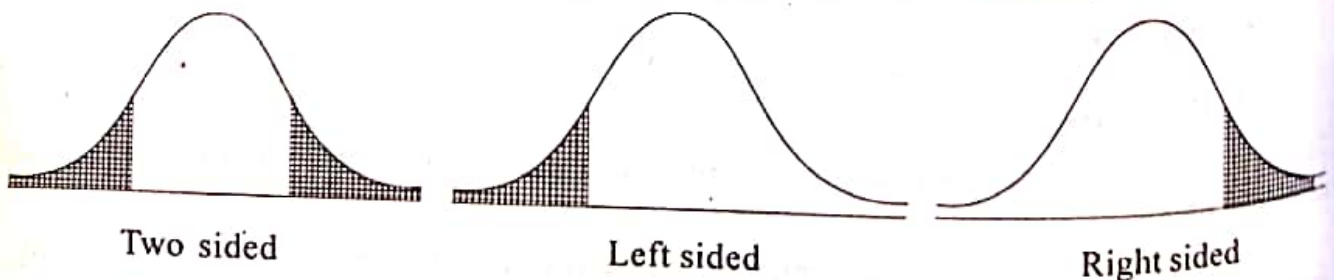
The level of significance denoted by α is the confidence with which we reject or accept the Null hypothesis H_0 i.e., it is the maximum possible probability with which we are willing to risk an error in rejecting H_0 when it is true. The level of significance is generally specified before a test procedure so that the results obtained may not influence our decision. In practice, we take either 5% (i.e., 0.05) or 1% (i.e., 0.01) level of significance, although other levels such as 2%, 1/2% etc. may also be used. 5% Level of significance in a test procedure indicates that there are about 5 cases in 100 that we would reject the null hypothesis H_0 when it is true i.e., we are about 95% confident that we have made the right decision. Similarly, in 1% Level of significance, there is only 1 case in 100 that the null hypothesis H_0 is rejected when it is true i.e., we are about 99% confident that we have made the right decision. Level of significance is also known as the size of the test.

Step 3 : Identification of the Test Statistic

There are several tests of significance, viz., z, t, F etc. First we have to select the right test depending on the nature of the information given in the problem. Then we construct the test criterion and select the appropriate probability distribution.

Step 4: Critical Region: The critical region is formed based on following factors.

- Distribution of the Statistic i.e., whether the statistic follows the normal, ' t ', χ^2 or ' F ' distribution (will be discussed later).
- Form of Alternative Hypothesis:** If the form has $\neq$ sign, the critical region is divided equally in the left and right tails, sides of the distribution.



If the form of alternative hypothesis has $<$ sign, the entire critical region is taken in the left tail of the distribution.

If the form of alternative hypothesis has $>$ sign, the entire critical region is taken on the right side of the distribution.

Step 5 : Making Decision

We compute the value of the appropriate statistic and ascertain whether the computed value falls in acceptance or rejection region depending on the specified Level of significance. In finding the acceptance or rejection region we have to use critical values given in Statistical Tables. By comparing the computed value and the critical value decision is taken for accepting or rejecting H_0 . If the computed value $<$ critical value, we accept H_0 , otherwise we reject H_0 .

9.13 ERRORS OF SAMPLING

[JNTU (H) Dec. 2009 (Set No. 2)]

The main objective in sampling theory is to draw valid inferences about the population parameters on the basis of the sample results. In practice we decide to accept or to reject the lot after examining a sample from it. As such we have two types of errors.

(i) **Type I error** : Reject H_0 when it is true. [JNTU (A) Dec. 2009 (Set No. 3)]

If the Null Hypothesis H_0 is true but it rejected by test procedure, then the error made is called Type I error or α error.

(ii) **Type II error** : Accept H_0 when it is wrong i.e., accept H_0 when H_1 is true. If the Null Hypothesis is false but it is accepted by test, then error committed is called Type II error or β error.

If we write

$$P(\text{Reject } H_0 \text{ when it is true}) = P(\text{Type I error}) = \alpha$$

$$\text{and } P(\text{Accept } H_0 \text{ when it is wrong}) = P(\text{Type II error}) = \beta$$

then α and β are called sizes of Type I and Type II errors respectively

$$\text{i.e., } \alpha = P(\text{Rejecting a good lot})$$

$$\beta = P(\text{Accepting a bad lot})$$

The sizes of Type I and Type II errors are also known as *producer's risk* and *consumer's risk* respectively.

The statistical testing of hypothesis aims at limiting the Type I error to a preassigned value (say : 1% or 5%) and to minimize the type II error. The only way to reduce both types of errors is to increase the sample size, if possible.

1. Critical Region

Under a given hypothesis let the sampling distribution of a statistic t is approximately a normal distribution with mean $E(t)$ and S. D. $= \sigma_t = \text{S. E. of } t$.

$$\text{Then } z = \frac{t - E(t)}{\text{S.E. of } t} = \frac{\text{Observed value} - \text{Expected value}}{\text{S.E. of } t} \text{ is called the standardized}$$

normal variate or test statistic or z -score and its distribution is the standard normal distribution with mean 0 and S.D. 1 when the sample is large.

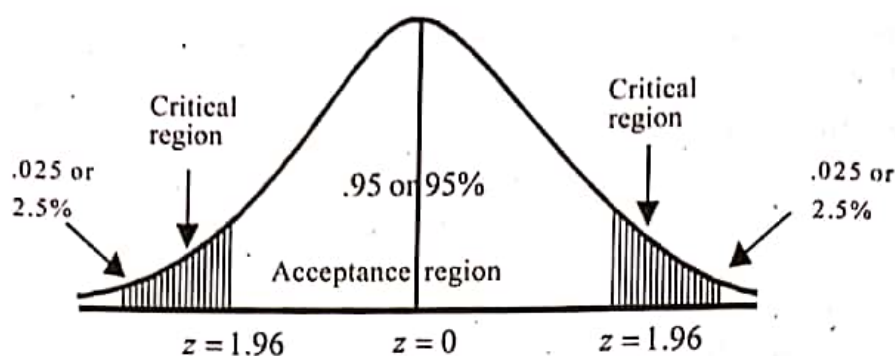


Fig.

From above figure, we see that if z lies between -1.96 and 1.96 then we are 95% confident that the hypothesis is true, since the area under the standard normal curve between $z = -1.96$ and $z = 1.96$ is 0.95 i.e., 95% of the total area i.e. $P(-1.96 \leq z \leq 1.96) = 0.95$. But if for a simple random sample we find that z lies outside the range -1.96 to 1.96 i.e., if $|z| > 1.96$, then we say that such an event occurs with probability of only 0.05 when the given hypothesis is true. In this case, we say that z -score differ significantly from the value expected under the hypothesis and hence the hypothesis H_0 is to be rejected at 5% level of significance. Thus if $|z| > 1.96$, the hypothesis H_0 is rejected at 5% level of significance. The set of z -scores outside the range -1.96 to 1.96 i.e., $|z| > 1.96$ constitutes the *critical region* or *region of rejection* of the hypothesis or the *region of significance*. On the otherhand, the set of z -scores inside the range -1.96 to 1.96 is called the region of acceptance of the hypothesis. The values -1.96 and 1.96 are called *critical values* at 5% level of significance.

Similarly, we can define critical region at any other level of significance.

Definition. A region corresponding to a statistic ' t ', in the sample space S which leads to the rejection of H_0 is called *Critical Region* or *Rejection Region*. Those region which lead to the acceptance of H_0 give us a region called *Acceptance Region*.

In other words, the region in which a sample value falling is rejected is known as the critical region. In general we take two critical regions which cover 5% and 1 % areas of the normal curve.

2. Critical Values or Significant Values :

[JNTU 2004S]

The value of the test statistic, which separates the critical region (or rejection region) and the acceptance region is called the *critical value* or *significant value*. This value is dependent on

- (i) the level of significance used, and
- (ii) the alternative hypothesis, whether it is one-tailed or two-tailed.

For larger samples, corresponding to the statistic t , the variable $z = \frac{t - E(t)}{\text{S.E of } t}$ is normally distributed with mean 0 and variance 1.

The value of z given above under the null hypothesis is known as test statistic. The critical value z_α of the test statistic at level of significance α for a two-tailed test is given by

$$P(|z| > z_\alpha) = \alpha \quad \dots (1)$$

That is, z_α is the value of Z so that the total area of the critical region on both tails is α . Since the normal curve is a symmetrical one, equation (1) implies,

$$P(z > z_\alpha) + P(z < -z_\alpha) = \alpha$$

$$\text{i.e. } 2 P(z > z_\alpha) = \alpha$$

$$\text{or } P(z > z_\alpha) = \alpha/2$$

That is, the area of each tail is $\alpha/2$.

The critical value z_α is that value such that the area to the right of z_α is $\alpha/2$ and the area to the left of $-z_\alpha$ is $\alpha/2$. (Refer figure).

3. Two-tailed test at level of significance ' α ' :

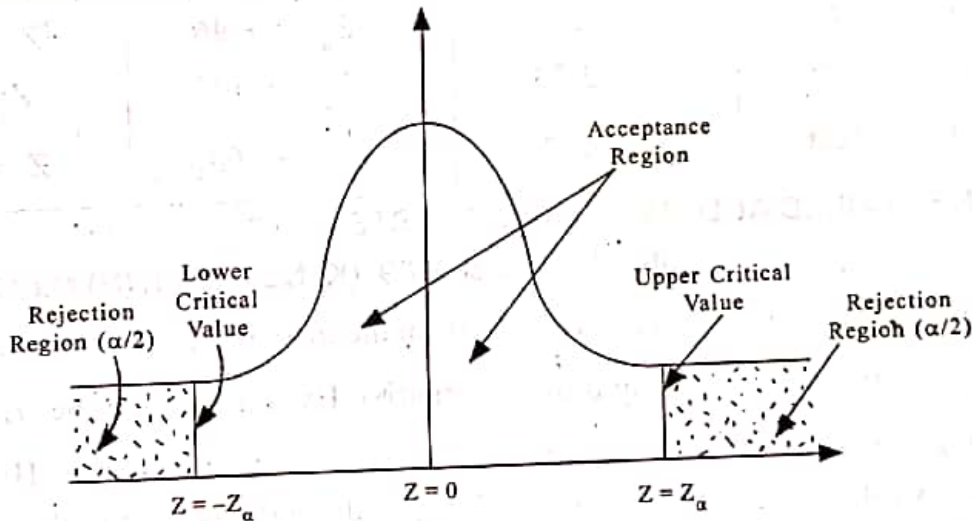


Fig.

In the case of one-tailed alternative,

$$P(Z > Z_\alpha) = \alpha \text{ if it is one-tailed (right)}$$

$$P(Z < -Z_\alpha) = \alpha \text{ if it is one-tailed (left)}$$

For Level of Significance ' α '

Right Tailed Test

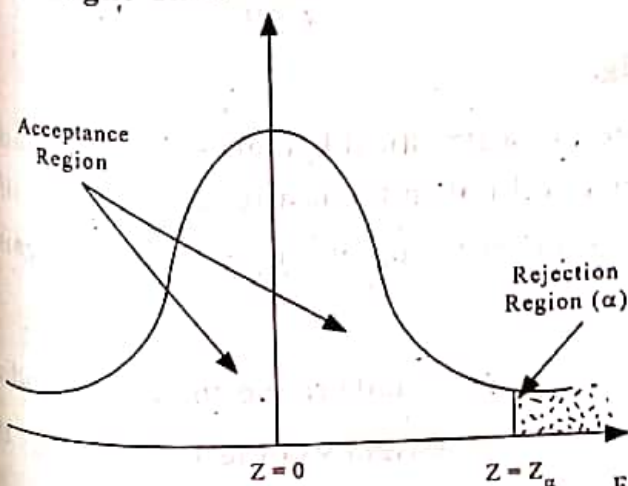
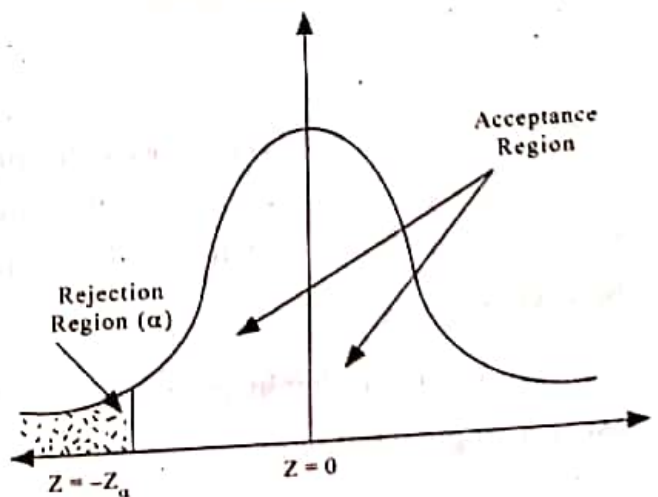


Fig.

Left Tailed Test



From the above figures, it is clear, that the critical value of Z for a single-tailed test (right or left) at level of significance ' α ' is same as the critical value of Z for two-tailed test at level of significance ' 2α '.

The critical values of Z at different level of significance (α) for both single tailed and two-tailed tests are calculated from equations

$$P(|Z| > Z_{\alpha}) = \alpha$$

$$P(Z > Z_{\alpha}) = \alpha$$

$$P(Z < -Z_{\alpha}) = \alpha$$

using the normal tables. They are listed below.

Critical values (Z_{α}) of Z

	Level of Significance		
	1% (.01)	5% (.05)	10% (.1)
Two-tailed test	$ Z_{\alpha} = 2.58$	$ Z_{\alpha} = 1.96$	$ Z_{\alpha} = 1.645$
Right-tailed test	$Z_{\alpha} = 2.33$	$Z_{\alpha} = 1.645$	$Z_{\alpha} = 1.28$
Left-tailed test	$Z_{\alpha} = -2.33$	$Z_{\alpha} = -1.645$	$Z_{\alpha} = -1.28$

9.14 ONE - TAILED AND TWO - TAILED TESTS

[JNTU (A) Dec. 2009, (K) Nov. 2011, (H) May 2012 (Set No. 1)]

If we have to test whether the population mean μ has a specified value μ_0 , then the Null Hypothesis is $H_0: \mu = \mu_0$ and the Alternative Hypothesis may be (i) $H_1: \mu \neq \mu_0$ (i.e., $\mu > \mu_0$ or $\mu < \mu_0$) or (ii) $H_1: \mu > \mu_0$ or (iii) $H_1: \mu < \mu_0$. The Alternative Hypothesis in (i) is known as a two-tailed (i.e., both right and left tail) alternatives and the alternative hypothesis in (ii) and (iii) are known as right - tailed and left - tailed alternatives respectively.

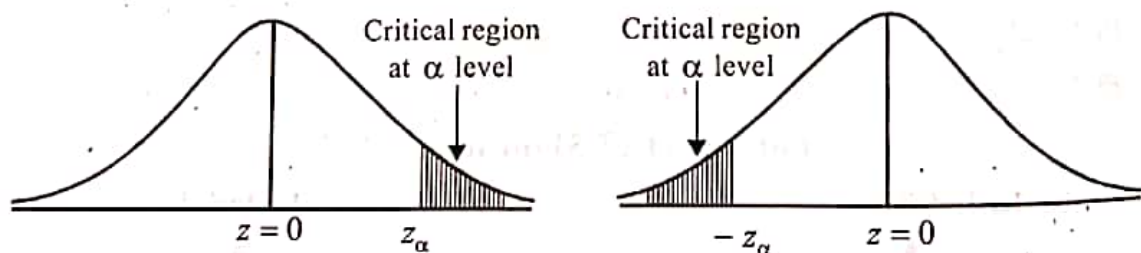


Fig.

If the Alternative Hypothesis H_1 in a test of a statistical hypothesis be one - tailed (i.e., either right - tailed or left - tailed but not both), then the test is called a *one-tailed test*. For example, to test whether the population mean $\mu = \mu_0$, we have $H_0: \mu = \mu_0$ against the alternative hypothesis H_1 given by

(i) $H_1: \mu > \mu_0$ (right tailed) or (ii) $H_1: \mu < \mu_0$ (left - tailed) and the corresponding test is a single - tailed or one-tailed or one sided. In the right - tail test $H_1: \mu > \mu_0$, the

critical region (or rejection region) $z > z_\alpha$ lies entirely in the right tail of the sampling distribution of sample mean $\bar{x}$ with area equal to the level of significance α (see figure). Similarly, in the left-tailed test ($H_1: \mu < \mu_0$), the critical region $z < -z_\alpha$ lies entirely in the left tail of the sampling distribution of the sample mean $\bar{x}$ with area equal to the level of significance α (see figure).

Two-Tailed Test :

Suppose we want to test the Null Hypothesis $H_0: \mu = \mu_0$ against the Alternative Hypothesis

$$H_1: \mu \neq \mu_0$$

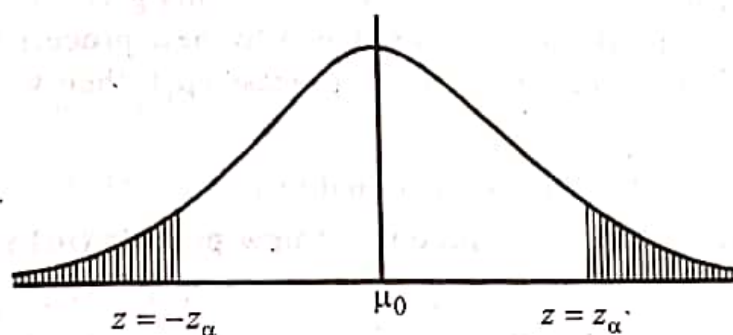


Fig.

Since H_1 is two-tailed alternative hypothesis, the critical region under the curve is equally distributed on both sides of the mean.

Thus, the critical area under the right-tail = The critical area under the left-tail

= Half of the total area

= $\frac{1}{2}$ probability of rejection

= $\frac{\alpha}{2}$ with critical statistic $Z_{\alpha/2}$,

where α is the level of significance.

The critical region is then, $z \leq -z_{\alpha/2}$ or $z_{\alpha/2} \leq z$.

Critical values of x for both two-tailed and one-tailed tests at 1%, 5% and 10% level of significance are given in the following table :

Table : Critical Values of z .			
Level of significance α	1%	5%	10%
Critical values for two-tailed test	$ z_\alpha = 2.58$	$ z_\alpha = 1.96$	$ z_\alpha = 1.645$
Critical values for Right-tailed test	$z_\alpha = 2.33$	$z_\alpha = 1.645$	$z_\alpha = 1.28$
Critical values for Left-tailed test	$z_\alpha = -2.33$	$z_\alpha = -1.645$	$z_\alpha = -1.28$

Applying one-tailed or two-tailed test for a particular problem depends entirely on the nature of the Alternative Hypothesis. If the alternative test is two-tailed we apply two-tailed test and if Alternative Hypothesis is one-tailed, we apply one-tailed test.

For example, to test whether a coin is biased or not, two - tailed test should be used, since a biased coin gives either more number of heads than tails (which corresponds to right tail) or more number of tails than heads (which corresponds to left tail)

Example : Consider two population brands of bulbs one manufactured by routine process (mean μ_1) and the other manufactured by new technique (mean μ_2). If we want to test if the bulbs differ significantly then the hypothesis is $H_0 : \mu_1 = \mu_2$ and the alternative hypothesis will be $H_0 : \mu_1 \neq \mu_2$. This gives us a two-tailed test. Suppose if we want to test if the bulbs produced by new process (μ_2) have higher average life than those produced by standard process (μ_1), then we have

$$H_0 : \mu_1 = \mu_2 \text{ and } H_1 : \mu_1 < \mu_2$$

In this case we have to adopt a left-tailed test.

If we want to test whether the product of new process (μ_2) is inferior to that of standard process (μ_1), then we have

$$H_0 : \mu_1 = \mu_2 \text{ and } H_1 : \mu_1 > \mu_2$$

which gives a right-tailed test.

Hence the decision about applying a two-tail test or a single-tail (left or right) test will mainly depend on the problem under study.

PROCEDURE FOR TESTING OF HYPOTHESIS :

[JNTU (H) Nov. 2009 (Set No.2,4), (K) Nov. 2011]

Various steps involved in testing of Hypothesis are given below : Infact, the same steps are followed for conducting all tests of significance.

Step 1 : Null Hypothesis : Define or set up a Null Hypothesis H_0 taking into consideration the nature of the problem and data involved.

Step 2 : Alternative Hypothesis : Set up the Alternative Hypothesis H_1 so that we could decide whether we should use one-tailed or two-tailed test.

Step 3 : Level of Significance : Select the appropriate level of significance(α) depending on the reliability of the estimates and permissible risk. That is, a suitable α is selected in advance if it is not given in the problem. (Usually we choose 5% level of significance)

Step 4 : Test Statistic: Compute the test statistic $Z = \frac{t - E(t)}{\text{S.E of } t}$ under the null hypothesis.

Here t is a sample statistic and S. E. is the standard error of t .

Step 5 : Conclusion : We compare the computed value of the test statistic Z with the critical value Z_α at given level of significance (α).

If $|Z| < Z_\alpha$, (that is, if the absolute value of the calculated value of Z is less than the critical value Z_α) we conclude that it is not significant. We accept the null hypothesis.

If $|Z| > Z_\alpha$ then the difference is significant and hence the null hypothesis is rejected at the level of significance α .

Clearly,

For Two-Tailed Test :If $|Z| < 1.96$ accept H_0 at 5% level of significance.If $|Z| > 1.96$ reject H_0 at 5% level of significance.If $|Z| < 2.58$, accept H_0 at 1% level of significance.If $|Z| > 2.58$ reject H_0 at 1% level of significance.**For Single-Tailed (Right or Left) Test :**If $|Z| < 1.645$, accept H_0 at 5% level of significance.If $|Z| > 1.645$, reject H_0 at 5% level of significance.If $|Z| < 2.33$ accept H_0 at 1% level of significance.If $|Z| > 2.33$ reject H_0 at 1% level of significance.**9.15 TEST OF SIGNIFICANCE FOR LARGE SAMPLES**

If the sample size $n \geq 30$, then we consider such samples as large samples. The tests of significance used in large samples are different from those used in small samples because small samples fail to satisfy the assumptions under which large sample analysis is done. If n is large, the distributions, such as Binomial, Poisson, Chi-square etc. are closely approximated by normal distributions. Therefore, for large samples, the sampling distribution of a statistic is approximately a normal distribution.

Suppose we wish to test the hypothesis that the probability of success in such trial is p . Assuming it to be true, the mean μ and the standard deviation σ of the sampling distribution of number of successes are np and $\sqrt{npq}$ respectively.

If x be the observed number of successes in the sample and Z is the standard normal variate then $Z = \frac{x - \mu}{\sigma}$.

Thus we have the following test of significance :

- (i) If $|Z| < 1.96$, the difference between the observed and expected number of successes is not significant.
- (ii) If $|Z| > 1.96$, the difference is significant at 5% level of significance.
- (iii) If $|Z| > 2.58$, the difference is significant at 1% level of significance.

Assumptions for Large Samples :

The following are the assumptions under which significance tests are applied :

1. The random sampling distribution of statistic has the properties of the normal curve. This may not hold good in case of small samples.
2. Values (i.e., statistic) given by the samples are sufficiently close to the population values (i.e., parameters) and can be used in its place for calculating the standard error (S. E.) of the estimate.

For example, for computing the S. E., if the S. D. of the universe is not known, it can be replaced by the S. D. of the sample. In case of small samples, this is not possible.

SOLVED EXAMPLES

Example 1 : A coin was tossed 960 times and returned heads 183 times. Test the hypothesis that the coin is unbiased. Use a 0.05 level of significance.

Solution : Here $n = 960$, $p = \text{Probability of getting head} = 1/2$

$$\therefore q = 1 - p = \frac{1}{2}; \quad \mu = np = 960 \left(\frac{1}{2} \right) = 480$$

$$\sigma = \sqrt{npq} = \sqrt{(np)q} = \sqrt{480 \times \frac{1}{2}} = \sqrt{240} = 15.49$$

$x = \text{number of successes} = 183$

1. Null Hypothesis H_0 : The coin is unbiased
2. Alternative Hypothesis H_1 : The coin is biased
3. Level of significance : $\alpha = 0.05$

$$4. \text{ The test statistic is } Z = \frac{x - \mu}{\sigma} = \frac{183 - 480}{15.49} = \frac{-297}{15.49} = -19.17$$

$$\therefore |Z| = 19.17$$

As $|Z| > 1.96$, the null hypothesis H_0 has to be rejected at 5% level of significance and we conclude that the coin is biased.

Example 2 : A coin was tossed 400 times and returned heads 216 times. Test the hypothesis that the coin is unbiased. Use a 0.05 Level of significance.

[JNTU (K) Nov. 2009 (Set No.3)]

Solution : Here $n = 400$

$$p = \text{Probability of getting head} = \frac{1}{2}$$

$$q = 1 - p = 1 - \frac{1}{2} = \frac{1}{2}, \quad \mu = np = 400 \left(\frac{1}{2} \right) = 200$$

$$\sigma = \sqrt{npq} = \sqrt{np(q)} = \sqrt{200 \times \frac{1}{2}} = \sqrt{100} = 10$$

$x = \text{number of successes} = 216$

1. Null Hypothesis H_0 : The coin is unbiased
2. Alternative Hypothesis H_1 : The coin is biased
3. Level of significance : $\alpha = 0.05$

$$4. \text{ The test statistic is } z = \frac{x - \mu}{\sigma}$$

$$\therefore z = \frac{216 - 200}{10} = \frac{16}{10} = 1.6$$

As $|z| < 1.96$, the null hypothesis H_0 has to be accepted and we conclude that the coin is unbiased.

Example 3 : A die is tossed 960 times and it falls with 5 upwards 184 times. Is the die unbiased at a level of significance of 0.01 ? [JNTU (K) Nov. 2009, May 2013 (Set No. 2)]

Solution : Here $n = 960$, $p =$ The probability of throwing 5 with one die $= 1/6$

$$\therefore q = 1 - \frac{1}{6} = \frac{5}{6}; \quad \mu = np = 960 \left(\frac{1}{6} \right) = 160$$

$$\sigma = \sqrt{npq} = \sqrt{160 \times \frac{5}{6}} = 11.55$$

$x =$ number of successes $= 184$

1. Null Hypothesis H_0 : The coin is unbiased
2. Alternative Hypothesis H_1 : The coin is biased
3. Level of significance : $\alpha = 0.01$

$$4. \text{ The test statistic is } Z = \frac{x - \mu}{\sigma} = \frac{184 - 160}{11.55} = \frac{24}{11.55} = 2.08$$

As $|Z| < 2.58$, the null hypothesis H_0 has to be accepted at 1 % level of significance and we conclude that the die is unbiased.

Note : As $|Z| > 1.96$, the null hypothesis H_0 has to be rejected at 5% level of significance and we conclude that the die is biased.

Example 4 : A die is tossed 256 times and it turns up with an even digit 150 times. Is the die biased ? [JNTU (K) Nov. 2011 (Set No. 3)]

Solution : Here $n = 256$, $p =$ The probability of getting an even digit (2 or 4 or 6) $= \frac{3}{6} = \frac{1}{2}$

$$q = 1 - p = \frac{1}{2}; \quad \mu = np = 256 \left(\frac{1}{2} \right) = 128$$

$$\sigma = \sqrt{npq} = \sqrt{np(q)} = \sqrt{128 \times \frac{1}{2}} = \sqrt{64} = 8$$

$x =$ number of successes $= 150$

1. Null Hypothesis H_0 : The die is unbiased
2. Alternative Hypothesis H_1 : The die is biased
3. Level of significance : $\alpha = 0.05$

$$4. \text{ The test statistic is } Z = \frac{x - \mu}{\sigma} \text{ i.e., } Z = \frac{150 - 128}{8} = \frac{22}{8} = 2.75$$

As $|Z| > 1.96$, the null hypothesis H_0 has to be rejected at 5% level of significance and we conclude that the die is biased.

Under large sample tests, we will see four important tests to test the significance.

1. Testing of significance for single proportion.
2. Testing of significance for difference of proportions.
3. Testing of significance for single mean.
4. Testing of significance for difference of means.

9.16 TEST OF SIGNIFICANCE OF A SINGLE MEAN - LARGE SAMPLES

Let a random sample of size n ($n \geq 30$) has the sample mean $\bar{x}$, and μ be the population mean. Also the population mean μ has a specified value μ_0 .

Working Rule :

1. **The Null Hypothesis** is $H_0: \bar{x} = \mu$ i.e., "there is no significance difference between the sample mean and population mean" or the sample has been drawn from the parent population.

2. **The Alternative Hypothesis** is

(i) $H_1: \bar{x} \neq \mu$ ($\mu \neq \mu_0$)

or (ii) $H_1: \bar{x} > \mu$ ($\mu > \mu_0$)

or (iii) $H_1: \bar{x} < \mu$ ($\mu < \mu_0$)

Since n is large, the sampling distribution of $\bar{x}$ is approximately normal.

3. **Level of Significance.** Set the level of significance α .

4. **The Test Statistic :**

We have the following two cases.

Case I : When the standard deviation σ of population is known.

In this case, standard Error of Mean, S. E. ($\bar{x}$) = $\frac{\sigma}{\sqrt{n}}$, where n = sample size,

σ = s.d. of population.

$\therefore$ The test statistic is

$$z = \frac{\bar{x} - \mu}{\text{S.E.}(\bar{x})} = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}, \text{ where } \mu \text{ is the sample mean.}$$

Case II : When the standard deviation σ of population is not known.

In this case, we take s , the s.d. of sample to compute the S. E. of mean

$$\therefore \text{S.E.}(\bar{x}) = \frac{s}{\sqrt{n}}$$

Hence the test statistic is $z = \frac{\bar{x} - \mu}{\text{S.E.}(\bar{x})} = \frac{\bar{x} - \mu}{s / \sqrt{n}}$

Find the critical value z_α of z at the level of significance α from the normal table.

Decision :

- (a) If $|z| < z_\alpha$ we accept the Null Hypothesis H_0 .
 (b) If $|z| > z_\alpha$, we reject the Null Hypothesis H_0 .

The rejection rule for $H_0: \bar{x} = \mu$ (or $\mu = \mu_0$) is given below :

Table : Critical Values of z			
Level of significance α	1%	5%	10%
Critical region for $\mu \neq \mu_0$	$ z > 2.58$	$ z > 1.96$	$ z > 1.645$
Critical region for $\mu > \mu_0$	$z > 2.33$	$z > 1.645$	$z > 1.28$
Critical region for $\mu < \mu_0$	$z < -2.33$	$z < -1.645$	$z < -1.28$

Note: 1. We reject Null Hypothesis H_0 when $|z| > 3$ without mentioning any level of significance.

The test statistic is, $z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$ where σ is the S.D of the population

If the population S.D is not known, then use the statistic

$$z = \frac{\bar{x} - \mu}{S / \sqrt{n}} \text{ where } S \text{ is the sample S.D.}$$

2. The values $\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}}$ are called 95% fiducial limits or confidence limits for the mean of the population corresponding to the given sample.

Similarly, $\bar{x} \pm 2.58 \frac{\sigma}{\sqrt{n}}$ (or) $[\bar{x} - 2.58 (\text{S.E. of } \bar{x}), \bar{x} + 2.58 (\text{S.E. of } \bar{x})]$ are called

99% confidence limits and $\bar{x} \pm 2.33 \frac{\sigma}{\sqrt{n}}$ are called 98% confidence limits.

SOLVED EXAMPLES

Example 1 : According to the norms established for a mechanical aptitude test, persons who are 18 years old have an average height of 73.2 with a standard deviation of 8.6. If 4 randomly selected persons of that age averaged 76.7, test the hypothesis $\mu = 73.2$ against the alternative hypothesis $\mu > 73.2$ at the 0.01 level of significance. [JNTU 2004, 2005S (Set No. 1)]

Solution : Given $n = 4$, $\mu = 73.2$, $\bar{x}$ = mean of the sample = 76.7
 and σ = S.D of population = 8.6

1. Null Hypothesis $H_0 : \mu = 73.2$
2. Alternative Hypothesis $H_1 : \mu > 73.2$ (Right-tailed test)
3. Level of significance : $\alpha = 99\%$ (or probability is 0.01)
4. The test statistic is $z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{76.7 - 73.2}{\frac{8.6}{\sqrt{4}}} = \frac{3.5}{4.3} = 0.814$

Tabulated value of z at 99% level of significance is 2.33.

Hence calculated $z <$ tabulated z

$\therefore$ The null hypothesis H_0 is accepted. That is, $\bar{x}$ and μ do not differ significantly.

Example 2 : A sample of 64 students have a mean weight of 70 kgs. Can this be regarded as a sample from a population with mean weight 56 kgs and standard deviation 25 kgs with level of significance 0.05? [JNTU 06, (A) Nov. 10 (Set No. 2), Dec. 2015]

Solution :

Given $\bar{x}$ = Mean of the sample = 70 kgs

μ = Mean of the population = 56 kgs

σ = S.D of population = 25 kgs

and n = Sample size = 64

1. Null Hypothesis H_0 : A sample of 64 students with mean weight of 70 kgs can be regarded as a sample from a population with mean weight 56 kgs and standard deviation 25 kgs.
2. Alternative Hypothesis H_1 : Sample cannot be regarded as one coming from the population.
3. Level of significance : $\alpha = 0.05$ (Given)

4. The test statistic is $z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{70 - 56}{\left(\frac{25}{\sqrt{64}}\right)} = 4.48$

5. The null hypothesis H_0 is rejected, since $|Z| > 1.645$

Note : The null hypothesis can be rejected even at 1% level of significance.

Example 3 : An oceanographer wants to check whether the depth of the ocean in a certain region is 57.4 fathoms, as had previously been recorded. What can he conclude at the 0.05 level of significance, if readings taken at 40 random locations in the given region yielded a mean of 59.1 fathoms with a standard deviation of 5.2 fathoms.

[JNTU 2003, 2003 S, 2004]

Solution: Given $n = 40$, $\bar{x} = 59.1$ and $\sigma = 5.2$

1. Null Hypothesis $H_0 : \mu = 57.4$
2. Alternative hypothesis $H_1 : \mu \neq 57.4$
3. Level of significance : $\alpha = 0.05$
4. The test statistic is $Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{59.1 - 57.4}{5.2/\sqrt{40}} = 2.067$

Tabulated value of Z at 5% level of significance is 1.96

Hence calculated $Z >$ tabulated Z .

$\therefore$ The null hypothesis H_0 is rejected.

Example 4 : In a random sample of 60 workers, the average time taken by them to get to work is 33.8 minutes with a standard deviation of 6.1 minutes. Can we reject the null hypothesis $\mu = 32.6$ minutes in favour of alternative null hypothesis $\mu > 32.6$ at $\alpha = 0.025$ level of significance. [JNTU 2005 (Set No. 1)]

Solution: Given $n = 60$, $\bar{x} = 33.8$, $\mu = 32.6$ and $\sigma = 6.1$

1. Null Hypothesis $H_0 : \mu = 32.6$
2. Alternative Hypothesis $H_1 : \mu > 32.6$
3. Level of significance : $\alpha = 0.025$
4. The test statistic is $Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{33.8 - 32.6}{6.1/\sqrt{60}} = \frac{1.2}{0.7875} = 1.5238$

Tabulated value of Z at 0.025 level of significance is 2.58.

Hence calculated $Z <$ tabulated Z

$\therefore$ The null hypothesis H_0 is accepted.

Example 5 : A sample of 900 members has a mean of 3.4 cms and S.D 2.61 cms. Is this sample has been taken from a large population of mean 3.25 cm and S.D 2.61 cms. If the population is normal and its mean is unknown find the 95% fiducial limits of true mean.

[JNTU (H) May 2011 (Set No. 1), (K) Nov. 2011 (Set No. 1)]

Solution :

Given $n = 900$, $\mu = 3.25$

$\bar{x} = 3.4$ cm, $\sigma = 2.61$

and $S = 2.61$

1. Null Hypothesis H_0 : Assume that the sample has been drawn from the population with mean $\mu = 3.25$.
2. Alternative Hypothesis $H_1 : \mu \neq 3.25$.

3. The test statistic is, $Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{3.4 - 3.25}{2.61/\sqrt{900}} = 1.724$

i.e., $Z = 1.724 < 1.96$

$\therefore$ We accept the null hypothesis H_0 .

i.e., The sample has been drawn from the population with mean $\mu = 3.25$.

95% confidence limits are given by

$$\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}} = 3.4 \pm 1.96 \times \frac{2.61}{\sqrt{900}} = 3.4 \pm 0.1705$$

i.e., 3.57 and 3.2295

Example 6 : A sample of 400 items is taken from a population whose standard deviation is 10. The mean of the sample is 40. Test whether the sample has come from a population with mean 38. Also calculate 95% confidence interval for the population.

[JNTU 2005, (H) Dec. 2011 (Set No. 1), (K) May 2013 (Set No. 4)]

Solution : Given $n = 400$, $\bar{x} = 40$, $\mu = 38$ and $\sigma = 10$

1. Null Hypothesis $H_0 : \mu = 38$
2. Alternative Hypothesis $H_1 : \mu \neq 38$
3. Level of significance : $\alpha = 0.05$

4. The test statistic is, $Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{40 - 38}{10/\sqrt{400}} = 4$ i.e., $Z = 4 > 1.96$

$\therefore$ We reject the null hypothesis H_0 .

i.e., The sample is not from the population whose mean is 38.

95% confidence interval is $\left(\bar{x} - 1.96 \frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96 \frac{\sigma}{\sqrt{n}} \right)$

i.e., $\left(40 - \frac{1.96(10)}{\sqrt{400}}, 40 + \frac{1.96(10)}{\sqrt{400}} \right)$

or $\left(40 - \frac{1.96}{20}, 40 + \frac{1.96}{20} \right)$ or $(40 - 0.098, 40 + 0.098)$

i.e., (39.902, 40.098)

Example 7 : (i) An ambulance service claims that it takes on the average less than 10 minutes to reach its destination in emergency calls. A sample of 36 calls has a mean of 11 minutes and the variance of 16 minutes. Test the claim at 0.05 level significance.

[JNTU 2005, (H) May 2012 (Set No. 4)]

(ii) An ambulance service claims that it takes on the average 8.9 minutes to reach its destination in emergency calls. To check this claim, the agency which licenses ambulance services has them timed on 50 emergency calls, getting a mean of 9.3 minutes with a standard deviation of 1.6 minutes. What can they conclude at the level of significance of 0.05 ?

[JNTU (A) June 2018]

Solution : (i) Given $n = 36$, $\bar{x} = 11$, $\mu = 10$ and $\sigma = \sqrt{16} = 4$

1. Null Hypothesis $H_0 : \mu = 10$

2. Alternative Hypothesis $H_1 : \mu < 10$

3. Level of significance : $\alpha = 0.05$

4. The test statistic is, $Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{11 - 10}{4/\sqrt{36}} = \frac{6}{4} = 1.5$

Tabulated value of Z at 5% level of significance is 1.645.

Hence calculated $Z < \text{tabulated } Z$

$\therefore$ We accept the null hypothesis H_0 .

(ii) This is left as an exercise to the reader.

Example 8 : It is claimed that a random sample of 49 tyres has a mean life of 15200 km. This sample was drawn from a population whose mean is 15150 kms and a standard deviation of 1200 km. Test the significance at 0.05 level.

[JNTU 2005, 2006S, (K) May 2013 (Set No. 2)]

Solution : Given $n = 49$, $\bar{x} = 15200$, $\mu = 15150$ and $\sigma = 1200$

1. Null Hypothesis $H_0 : \mu = 15150$

2. Alternative Hypothesis $H_1 : \mu \neq 15150$

3. Level of significance : $\alpha = 0.05$

4. Critical region : Accept the null hypothesis if $-1.96 < Z < 1.96$

5. The test statistic is, $Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{15200 - 15150}{1200/\sqrt{49}} = 0.2917$

Since $|Z| < 1.96$ therefore, we accept the null hypothesis.

Example 9 : An insurance agent has claimed that the average age of policy holders who issue through him is less than the average for all agents which is 30.5 years. A random sample of 100 policy holders who had issued through him gave the following age distribution.

Age	16-20	21-25	26-30	31-35	36-40
No. of persons	12	22	20	30	16

Calculate the Arithmetic mean and Standard deviation of this distribution and use these values to test his claim at 5% level of significance.

Solution : Take $A = 28$, $d_i = x_i - A$

$$\therefore \text{A.M.} = \bar{x} = A + \frac{h \sum f_i d_i}{N} = 28 + \frac{9 \times 16}{100} = 28.8$$

$$\begin{aligned} \text{S.D. : } S &= h \sqrt{\frac{\sum f d^2}{N} - \left(\frac{\sum f d}{N}\right)^2} = 5 \cdot \sqrt{\frac{164}{100} - \left(\frac{16}{100}\right)^2} \quad [\because h = 5] \\ &= 6.35 \end{aligned}$$

1. **Null Hypothesis H_0 :** The sample is drawn from a population with mean μ i.e. $\bar{x}$ and μ do not differ significantly where $\mu = 30.5$ years.
2. **Alternative Hypothesis H_1 :** $\mu < 30.5$ years (left tail test)

Now, $\bar{x} = 28.8$, $S = 6.35$, $\mu = 30.5$ years and $n = 100$

$$3. \text{ The test statistic is, } Z = \frac{\bar{x} - \mu}{S/\sqrt{n}} = \frac{28.8 - 30.5}{6.35/\sqrt{100}} = -2.677$$

$$\therefore |Z| = 2.68.$$

Tabulated value of Z at 5% level of significance is 1.645 (left tail test).

Here calculated $Z >$ tabulated Z .

$\therefore$ The Null hypothesis H_0 is rejected.

i.e., $\bar{x}$ and μ differ significantly.

i.e., The sample is not drawn from a population with mean $\mu = 30.5$ years

Example 10 : The mean life time of a sample of 100 light tubes produced by a company is found to be 1560 hrs with a population S.D of 90 hrs. Test the hypothesis for $\alpha = 0.05$ that the mean life time of the tubes produced by the company is 1580 hrs.

[JNTU (A) Dec. 2009 (Set No. I)]

Solution : Given $\bar{x}$ = Mean of the sample = 1560 hrs

μ = Mean of the population = 1580 hrs

n = Sample size = 100

σ = Standard deviation = 90 hrs

1. **Null Hypothesis H_0 :** $\mu = 1580$
2. **Alternative Hypothesis H_1 :** $\mu \neq 1580$

3. **Level of significance :** $\alpha = 0.05$

$$4. \text{ The test statistic is } z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{1560 - 1580}{90/\sqrt{100}} = \frac{-20}{9}$$

$$\therefore |z| = \frac{20}{9} = 2.22$$

$z_{\alpha/2}$ for $\alpha = 0.05$ is 1.96. Since $|z| > 1.96$, the null Hypothesis H_0 is rejected.
 $\therefore \mu \neq 1580$.

Note: $z_{\alpha/2}$ for $\alpha = 0.01$ is 2.58

Since $|z| < 2.58$, the null Hypothesis H_0 can be accepted at 1% level of significance.

Example 11 : If the distribution of the weights of all men travelling by air between Delhi and Mumbai has a mean of 163 pounds and a standard deviation of 18 pounds. What is the probability that the consigned gross weight of 36 men travelling between these two cities is more than 6000 pounds.

[JNTU (A) Dec. 2009 (Set No. 2)]

Solution : Given μ = The mean of the population = 163 pounds

σ = The standard deviation of the population = 18 pounds

Gross weight of 36 men = 6000 pounds

$$\therefore \bar{x} = \text{Average weight of a man} = \frac{6000}{36} = 166.7 \text{ pounds}$$

Also we have n = sample size = 36

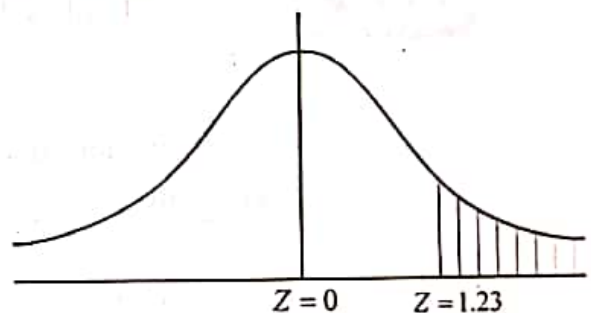
$$\therefore z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{166.7 - 163}{18 / \sqrt{36}} = \frac{3.7}{3} = 1.23$$

$$\text{Hence } P(\bar{x} \geq 166.7) = P(Z \geq 1.23)$$

$$= 0.5 - P(Z < 1.23)$$

$$= 0.5 - 0.3907 \text{ (from normal tables)}$$

$$= 0.1093$$



Thus the probability that the consigned gross weight of 36 men travelling between these two cities is more than 6000 pounds is 0.1093.

Example 12 : The length of life X of certain computers is approximately normally distributed with mean 800 hours and standard deviation 40 hours. If a random sample of 30 computers has an average life of 788 hours, test the null hypothesis that $\mu = 800$ hours against the alternative that $\mu \neq 800$ hours at (i) 0.5% (ii) 1% (iii) 4% (iv) 5% (v) 10% (vi) 15% level.

[JNTU (K) May 2010 (Set No. 1)]

Solution : The Null Hypothesis is $H_0 = 800$ hours.

The Alternative Hypothesis is $H_1 : \mu \neq 800$ hours.

Since H_1 is two-tailed (i.e. two sided), we are to use two -tailed test.

Let us assume that H_0 is true.

$$\text{The test statistic is } z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

Here $\bar{x}$ = sample mean = 788 hours

$n = \text{sample size} = 30$

$\sigma = \text{standard deviation} = 40$

$$\therefore z = \frac{788 - 800}{40 / \sqrt{30}} = -1.643$$

and $|z| = 1.643$

- (i) Since $|z| < 2.81$, we accept the Null Hypothesis H_0 at 0.5% level of significance.
- (ii) Since $|z| < 2.58$, we accept the Null Hypothesis H_0 at 1% level of significance.
- (iii) Since, $|z| < 2.06$, we accept H_0 at 4% level of significance.
- (iv) Since, $|z| < 1.96$, we accept H_0 at 5% level of significance.
- (v) Since, $|z| < 1.645$, we accept H_0 at 10% level of significance.
- (vi) Since, $|z| > 1.44$, we reject H_0 at 15% level of significance.

Example 13 : In 64 randomly selected hours of production, the mean and the standard deviation of the number of acceptance pieces produced by an automatic stamping machine are $\bar{x} = 1.038$ and $\sigma = .146$

At the .05 level of significance does this enable us to reject the null hypothesis $\mu = 1.000$ against the alternative hypothesis $\mu > 1.000$? [JNTU (H) Nov. 2010 (Set No. 4)]

Solution : Let the Null Hypothesis be $H_0 : \mu = 1.000$

Then the Alternative Hypothesis is $H_1 : \mu > 1.000$

Here $\bar{x} = \text{Mean of the sample} = 1.038$

$\mu = \text{Mean of the population} = 1.000$

$\sigma = \text{S.D. of the population} = 0.146$

and $n = \text{Sample size} = 64$

$$\begin{aligned} \therefore \text{The test statistic is } z &= \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} \\ &= \frac{1.038 - 1.000}{0.146 / \sqrt{64}} = \frac{0.038}{0.146 / 8} = 2.082 \end{aligned}$$

Thus we see that $z = 2.082 > 1.645$

Hence, we reject the Null Hypothesis H_0 at 5% level of significance and conclude that the mean of the population $\mu > 1.000$

Example 14 : A trucking rm suspects the claim that average life of certain tyres is atleast 28,000 miles. To check the claim the rm puts 40 of these tyres on its trucks and gets a mean life time of 27463 miles with a standard deviation of 1348 miles. Can the claim be true?

[JNTU (H) Apr. 2012 (Set No. 2)]

Solution : The Null Hypothesis is $H_0 : \mu = 28,000$ miles
The Alternative Hypothesis is $H_1 : \mu \neq 28,000$

Since H_1 is two - tailed (i.e., two - sided), we are to use two - tail test.
Let us assume that H_0 is true.

The test statistic is $z = \frac{\bar{x} - \mu}{\text{S.E. of } \bar{x}} = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$

Here $\bar{x}$ = sample mean = 27463 miles, $n = 40$ and $\sigma = s = 1348$ miles
[$\therefore$ population S. D. σ is not known]

$$\therefore z = \frac{27463 - 28000}{1348 / \sqrt{40}} = \frac{-537(\sqrt{40})}{1348} = -2.52$$

$$\text{and } |z| = |-2.52| = 2.52 > 1.96$$

Hence, the Null Hypothesis H_0 is rejected at 5% level of significance and we conclude that the mean life of tyres, cannot be taken as 28000 miles.

9.17 TEST FOR EQUALITY OF TWO MEANS - LARGE SAMPLES (TEST OF SIGNIFICANCE FOR DIFFERENCE OF MEANS OF TWO LARGE SAMPLES)

Let $\bar{x}_1$ and $\bar{x}_2$ be the sample means of two independent large random samples sizes n_1 and n_2 drawn from two populations having means μ_1 and μ_2 and standard deviations σ_1 and σ_2 . To test whether the two population means are equal.

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$

Then the Alternative Hypothesis is $H_1 : \mu_1 \neq \mu_2$

S. E. of $(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$, where σ_1 and σ_2 are the S. D. of the two populations

To test whether there is any significant difference between $\bar{x}_1$ and $\bar{x}_2$, we have to use the statistic

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - \delta}{\text{S.E. of } (\bar{x}_1 - \bar{x}_2)} = \frac{(\bar{x}_1 - \bar{x}_2) - \delta}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

where $\delta = \mu_1 - \mu_2$ (= given constant.)

If $\delta = 0$, the two populations have the same means.

If $\delta \neq 0$ the two populations are different.

Under $H_0: \mu_1 = \mu_2$, the test statistic becomes
$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

is approximately normally distributed with mean 0 and S. D. 1.

Note : If the samples have been drawn from the population with common S. D. σ , then $\sigma_1^2 = \sigma_2^2 = \sigma^2$

$$\text{Hence } z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}}} = \frac{\bar{x}_1 - \bar{x}_2}{\sigma \cdot \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

is normally distributed with mean zero and standard deviation one.

If σ is not known we can use an estimate of σ^2 given by $\sigma^2 = \frac{n_1 S_1^2 + n_2 S_2^2}{n_1 + n_2}$

Rejection Rule for $H_0: \mu_1 = \mu_2$

- (i) If $|z| > 1.96$, reject H_0 at 5% level of significance.
- (ii) If $|z| > 2.58$, reject H_0 at 1% level of significance.
- (iii) If $|z| > 1.645$, reject H_0 at 10% level of significance.
- (iv) If $z > 3$ then either the samples have not been drawn from the same population or the sampling is not simple.

Otherwise accept H_0 .

Note : If the two samples are drawn from two populations with unknown Standard deviations σ_1^2 and σ_2^2 , then σ_1^2 and σ_2^2 can be replaced by sample variances S_1^2 and S_2^2 provided both the samples n_1 and n_2 are large.

In this case, the test statistic is
$$z = \frac{(\bar{x}_1 - \bar{x}_2) - \delta}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

Write the formula for testing the hypothesis concerning "Two Means".

[JNTU Nov. 2008 (Set No.1)]

Solution : Let $\bar{x}_1$ be the mean of a random sample of size n_1 drawn from a population with mean μ_1 and variance σ_1^2 . Let $\bar{x}_2$ be the mean of an independent random sample of size n_2 drawn from another population with mean μ_2 and variance σ_2^2 . To test the hypothesis for difference of means, the statistic is given by

$$z = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

If $|z| < 1.96$, it is accepted that the samples have come from two populations with the same mean, at 5% level of significance. Otherwise, at this level of significance, we claim that the difference in means is significant.

SOLVED EXAMPLES

Example 1 : The means of two large samples of sizes 1000 and 2000 members are 67.5 inches and 68.0 inches respectively. Can the samples be regarded as drawn from the same population of S.D 2.5 inches. [JNTU (A) Nov. 2010 (Set No. 1)]

Solution : Let μ_1 and μ_2 be the means of the two populations.

Given $n_1 = 1000$, $n_2 = 2000$ and $\bar{x}_1 = 67.5$ inches, $\bar{x}_2 = 68$ inches

Population S.D, $\sigma = 2.5$ inches

1. **Null Hypothesis H_0 :** The samples have been drawn from the same population of S.D 2.5 inches

i.e., $\mu_1 = \mu_2$ and $\sigma = 2.5$ inches

2. **Alternative Hypothesis H_1 :** $\mu_1 \neq \mu_2$

3. **The test statistic is,**
$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}}} = \frac{67.5 - 68}{\sqrt{(2.5)^2 \left(\frac{1}{1000} + \frac{1}{2000} \right)}}$$

$$\Rightarrow z = \frac{-0.5}{0.0968} = -5.16$$

$\therefore |z| = 5.16 > 1.96$ i.e., the calculated value of $z >$ the table value of z .

Hence the null hypothesis H_0 is rejected at 5% level of significance and we conclude that the samples are not drawn from the same population of S.D. 2.5 inches.

Example 2 : The mean yield of wheat from a district A was 210 pounds with S.D. 10 pounds per acre from a sample of 100 plots. In another district the mean yield was 220 pounds with S.D. 12 pounds from a sample of 150 plots. Assuming that the S.D of yield in the entire state was 11 pounds, test whether there is any significant difference between the mean yield of crops in the two districts.

Solution : Let μ_1 and μ_2 be the means of the two populations.

Given $\bar{x}_1 = 210$, $\bar{x}_2 = 200$, and $n_1 = 100$, $n_2 = 150$,

Population S.D., $\sigma = 11$

1. **Null Hypothesis H_0 :** $\mu_1 = \mu_2$ i.e. there is no difference between μ_1 and μ_2

2. **Alternative Hypothesis H_1 :** $\mu_1 \neq \mu_2$

3. **The test statistic is**
$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}}} = \frac{210 - 200}{\sqrt{11^2 \left(\frac{1}{100} + \frac{1}{150} \right)}} = 7.04178$$

$\therefore |z| = 7.041 > 1.96$ i.e. the calculated value of $z >$ the table value of z .

Hence we reject the Null Hypothesis H_0 at 5% level of significance and conclude that there is a significant difference between the mean yield of crops in the two districts.

Example 3 : In a survey of buying habits, 400 women shoppers are chosen at random in super market 'A' located in a certain section of the city. Their average weekly food expenditure is Rs. 250 with a S.D. of Rs. 40. For 400 women shoppers chosen at random in super market 'B' in another section of the city, the average weekly food expenditure is Rs. 220 with a S.D. of Rs. 55. Test at 1% level of significance whether the average weekly food expenditure of the two populations of shoppers are equal.

Solution : Let μ_1 and μ_2 be the means of the two populations.

Given $n_1 = 400$, $\bar{x}_1 = \text{Rs. } 250$, $S_1 = \text{Rs. } 40$

and $n_2 = 400$, $\bar{x}_2 = \text{Rs. } 220$, $S_2 = \text{Rs. } 55$

1. **Null Hypothesis H_0 :** Assume that the average weekly food expenditure of the two populations of shoppers are equal i.e., $H_0 : \mu_1 = \mu_2$
2. **Alternative Hypothesis H_1 :** $\mu_1 \neq \mu_2$

$$3. \text{ Test statistic is, } z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} = \frac{250 - 220}{\sqrt{\frac{(40)^2}{400} + \frac{(55)^2}{400}}} = \frac{30}{3.4} = 8.82$$

i.e. $z = 8.82 > 2.58$

Hence we reject the Null Hypothesis H_0 at 1% level of significance and conclude that the average weekly food expenditure of the two populations of shoppers are not equal.

Example 4 : Samples of students were drawn from two universities and from their weights in kilograms, mean and standard deviations are calculated and shown below. Make a large sample test to test the significance of the difference between the means.

	Mean	S.D	Size of the sample
University A	55	10	400
University B	57	15	100

[JNTU 2005 (Set No. 3)]

Solution :

Given $\bar{x}_1 = 55$, $\bar{x}_2 = 57$, $n_1 = 400$, $n_2 = 100$, $S_1 = 10$ and $S_2 = 15$

1. **Null Hypothesis H_0 :** $\bar{x}_1 = \bar{x}_2$ i.e., there is no difference
2. **Alternative Hypothesis H_1 :** $\bar{x}_1 \neq \bar{x}_2$
3. **Level of significance :** $\alpha = 0.05$ (assumed)
4. **Critical region :** Accept H_0 if $-1.96 < Z < 1.96$

$$5. \text{ The test statistic is } Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} = \frac{55 - 57}{\sqrt{\frac{100}{400} + \frac{225}{100}}} = \frac{-2}{\sqrt{\frac{1}{4} + \frac{9}{4}}} = -1.26$$

$$\therefore |Z| = 1.26 < 1.96$$

Hence, we accept the Null Hypothesis H_0 at 5% level of significance and conclude that there is no significant difference between the means.

Example 5 : The research investigator is interested in studying whether there is a significant difference in the salaries of MBA grades in two metropolitan cities. A random sample of size 100 from Mumbai yields on average income of Rs. 20,150. Another random sample of 60 from Chennai results in an average income of Rs. 20,250. If the variances of both the populations are given as $\sigma_1^2 = \text{Rs. } 40,000$ and $\sigma_2^2 = \text{Rs. } 32,400$ respectively.

[JNTU Nov. 2008 (Set No. 1)]

Solution : Given $n_1 = 100$, $\bar{x}_1 = 20,150$ and $n_2 = 60$, $\bar{x}_2 = 20,250$

and $\sigma_1^2 = 40,000$, $\sigma_2^2 = 32,400$

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$. That is, the difference of means is significant.

Then the Alternative Hypothesis is $H_1 : \mu_1 \neq \mu_2$

$$\begin{aligned} \text{The test statistic is } z &= \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{20150 - 20250}{\sqrt{\frac{40000}{100} + \frac{32400}{60}}} \\ &= \frac{100}{\sqrt{400 + 540}} = \frac{100}{30.66} = 3.26 \end{aligned}$$

Since $Z = 3.26 > 1.96 = Z_{0.05}$, we reject the Null Hypothesis at 5% level of significance and conclude that there is a significant difference in the salaries of MBA grades in two cities.

Example 6 : A researcher wants to know the intelligence of students in a school. He selected two groups of students. In the first group there 150 students having mean IQ of 75 with a S.D. of 15 in the second group there are 250 students having mean IQ of 70 with S.D. of 20. Is there a significant difference between the means of two groups ?

[JNTU Nov. 2008 (Set No. 4)]

Solution : Given $n_1 = 150$, $\bar{x}_1 = 75$, $\sigma_1 = 15$

and $n_2 = 250$, $\bar{x}_2 = 70$, $\sigma_2 = 20$

1. Null Hypothesis H_0 : The groups have been came from the same population

i.e., $\mu_1 = \mu_2$

2. Alternative Hypothes is $H_1 : \mu_1 \neq \mu_2$

$$\begin{aligned} \text{3. The test statistic is, } z &= \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{75 - 70}{\sqrt{\frac{225}{150} + \frac{400}{250}}} \\ &= \frac{5}{\sqrt{\frac{9}{5} + \frac{8}{5}}} = \frac{5\sqrt{5}}{\sqrt{17}} = 2.7116 \end{aligned}$$

Tabulated value of z at 1% level of significance is 2.33

$\therefore$ Calculated $z >$ tabulated z .

Hence we reject the Null Hypothesis H_0 at 1% level of significance and conclude that the groups have not been taken from the same population.

Example 7 : Two types of new cars produced in U.S.A. are tested for petrol mileage, one sample is consisting of 42 cars averaged 15 kmpl while the other sample consisting of 80 cars averaged 11.5 kmpl with population variances as $\sigma_1^2 = 2.0$ and $\sigma_2^2 = 1.5$ respectively. Test whether there is any significance difference in the petrol consumption of these two types of cars. (use $\alpha = 0.01$). [JNTU Apr. 2009 (Set No. 1)]

Solution : Let the types of the cars be named as A and B.

Number of cars of type A = 42

Average mileage for A = $\bar{x}_1 = 15$, Variance = $\sigma_1^2 = 2.0$

Number of cars of type B = 80

Average mileage for B = $\bar{x}_2 = 11.5$

Variance = $\sigma_2^2 = 1.5$

1. Null Hypothesis $H_0 : \mu_1 = \mu_2$

2. Alternative Hypothesis is $H_1 : \mu_1 \neq \mu_2$

$$\begin{aligned} 3. \text{ The test statistic is } z &= \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{|15 - 11.5|}{\sqrt{\frac{2}{42} + \frac{1.5}{80}}} \\ &= \frac{3.5}{\sqrt{0.0476 + 0.01875}} = \frac{3.5}{\sqrt{0.06635}} = 13.587 \end{aligned}$$

Tabulated value of z at 1% significance level is 2.58 (two-tailed)

Since $(z)_{\text{calculated}} > 2.58$ (z table), we reject Null Hypothesis H_0 at 1% level of significance and conclude that there is a significant difference in petrol consumption.

Example 8 : A simple sample of the height of 6400 Englishmen has a mean of 67.85 inches and a S. D of 2.56 inches while a simple sample of heights of 1600 Austrains has a mean of 68.55 inches and S. D. of 2.52 inches. Do the data indicate the Austrains are on the average taller than the Englishmen? (Use α as 0.01). [JNTU (A) 2009 (Set No. 2)]

Solution : We are given

n_1 = Size of the first sample = 6400

n_2 = Size of the second sample = 1600

$\bar{x}_1$ = Mean of the first sample = 67.85

$\bar{x}_2$ = Mean of the second sample = 68.55

σ_1 = Standard deviation of the first sample = 2.56

σ_2 = Standard deviation of the second sample = 2.52

1. Null Hypothesis $H_0 : \mu_1 = \mu_2$
2. Alternative Hypothesis $H_1 : \mu_1 < \mu_2$
3. Level of significance : $\alpha = 0.05$
4. The test statistic is

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\left(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)}} = \frac{67.85 - 68.55}{\sqrt{\frac{(2.56)^2}{6400} + \frac{(2.52)^2}{1600}}} = \frac{-0.7}{\sqrt{\frac{6.5536}{6400} + \frac{6.35}{1600}}}$$

$$= \frac{-0.7}{\sqrt{0.001 + 0.004}} = \frac{-0.7}{0.0707} = -9.9$$

$$\therefore |z| = 9.9 > 1.96$$

Hence, we reject the Null Hypothesis H_0 at 5% level of significance and conclude that the Australians are taller than Englishmen.

Example 9 : The mean life of a sample of 10 electric bulbs (or motors) was found to be 1456 hours with S.D. of 423 hours. A second sample of 17 bulbs (motors) chosen from a different batch showed a mean life of 1280 hours with S.D. of 398 hours. Is there a significant difference between the means of two batches? [JNTU (K) 2009, Nov. 2012 (Set No. 1)]

Solution : It is given that

n_1 = Sample size of first batch = 10

n_2 = Sample size of second batch = 17

$\bar{x}_1$ = Mean life of first batch = 1456

$\bar{x}_2$ = Mean life of second batch = 1280

σ_1 = Standard deviation of first batch = 423

σ_2 = Standard deviation of second batch = 398

1. Null Hypothesis $H_0 : \mu_1 = \mu_2$
2. Alternative Hypothesis $H_1 : \mu_1 \neq \mu_2$
3. Level of significance : $\alpha = 0.05$

4. The test statistic is

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{1456 - 1280}{\sqrt{\frac{(423)^2}{10} + \frac{(398)^2}{17}}}$$

$$= \frac{176}{\sqrt{17892.9 + 9317.88}} = \frac{176}{164.96} = 1.067$$

Since $z < z_{\alpha} = 1.96$, we accept the null hypothesis H_0 i.e., there is no difference between the mean life of electric bulbs of two batches.

Example 10 : The average marks scored by 32 boys is 72 with a S.D. of 8. While that for 36 girls is 70 with a S.D. of 6. Does this indicate that the boys perform better than girls at level of significance 0.05?

[JNTU (K) May 2010 (Set No. 3)]

Solution : Let μ_1 and μ_2 be the means of the two populations

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$

Then the Alternative Hypothesis is $H_1 : \mu_1 > \mu_2$

Let us assume that H_0 is true, i.e., there is no difference between μ_1 and μ_2

Since the sample sizes are large we use the test statistic

$$z = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Here $\bar{x} = 72, \bar{y} = 70, \sigma_1 = 8, \sigma_2 = 6, n_1 = 32, n_2 = 36$

$$\therefore z = \frac{72 - 70}{\sqrt{\frac{64}{32} + \frac{36}{36}}} = \frac{2}{\sqrt{2+1}} = \frac{2}{\sqrt{3}} = 1.1547 < 1.96$$

Since the computed value of z is less than the table value, we cannot reject the Null Hypothesis at 5% level and conclude that the performance of boys and girls is the same.

Example 11 : At a certain large university a sociologist speculates that male students spend considerably more money on junk food than do female students. To test her hypothesis, the sociologist randomly selects from the registrar's records the names of 200 students. Of these, 125 are men and 75 are women. The sample mean of the average amount spent on junk food per week by the men is Rs. 400 and standard deviation is 100. For the women the sample mean is Rs. 450 and the sample standard deviation is Rs.150. Test the difference between the mean at .05 level.

[JNTU (H) Nov. 2010 (Set No. 2)]

Solution : Let μ_1 and μ_2 be the means of the two populations.

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$

Then the Alternative Hypothesis is $H_1 : \mu_1 \neq \mu_2$

Let us assume that H_0 is true i.e., there is no difference between μ_1 and μ_2 .

We are given .

n_1 = Number of men = 125

n_2 = Number of women = 75

$\bar{x}_1$ = Mean of men = 400

$\bar{x}_2$ = Mean of women = 450

$$\sigma_1 = \text{S.D. of men} = 100.$$

$$\sigma_2 = \text{S.D. of women} = 150$$

$$\text{Level of significance, } \alpha = 0.05$$

$$\begin{aligned} \text{The test statistic is } z &= \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{400 - 450}{\sqrt{\frac{(100)^2}{125} + \frac{(150)^2}{75}}} \\ &= \frac{-50}{\sqrt{80 + 300}} = \frac{-50}{\sqrt{380}} = \frac{-50}{19.49} = -2.5654 \end{aligned}$$

$\therefore |z| = |-2.5654| = 2.5654 > 1.96$ i.e., the difference is highly significant.

Hence, we reject the Null Hypothesis at 5% level of significance and conclude that the two population means are not equal.

Example 12 : A company claims that its bulbs are superior to those of its main competitor. If a study showed that a sample of 40 of its bulbs have a mean life time of 647 hrs of continuous use with a S.D of 27 hrs. While a sample of 40 bulbs made by its main competitor had a mean life time of 638 hrs of continuous use with a S.D of 31 hrs. Test the significance between the difference of two means at 5% level.

[JNTU (H) Nov. 2010 (Set No. 4)]

Solution : Let μ_1 and μ_2 be the means of the two populations.

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$

Then the Alternative Hypothesis is $H_1 : \mu_1 > \mu_2$

Since the sample sizes are large, the test statistic

$$z = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

is approximately normally distributed with mean 0 and S.D. 1.

Here $\bar{x} = 647, \bar{y} = 638, \sigma_1 = 27, \sigma_2 = 31, n_1 = n_2 = 40$

$$\begin{aligned} \therefore z &= \frac{647 - 638}{\sqrt{\frac{(27)^2}{40} + \frac{(31)^2}{40}}} \\ &= \frac{9}{\sqrt{\frac{729 + 961}{40}}} = \frac{9}{6.5} = 1.38 \end{aligned}$$

i.e., $|z| = 1.38 < 1.645$

Since the computed value of z is less than the table value, we cannot reject the Null Hypothesis at 5% level and conclude that the difference between the two sample means is not significant.

Example 13 : Studying the flow of traffic at two busy intersections between 4 p.m and 6 p.m to determine the possible need for turn signals. It was found that on 40 week days there were on the average 247.3 cars approaching the first intersection from the south which made left turn, while on 30 week days there were on the average 254.1 cars approaching the first intersection from the south made left turns. The corresponding sample standard deviations are 15.2 and 12. Test the significance between the difference of two means at 5% level.

[JNTU (H) Dec. 2011 (Set No.1)]

Solution : Let the average cars in two places be μ_1 and μ_2 respectively.

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$

Then the Alternative Hypothesis is $H_1 : \mu_1 \neq \mu_2$

Let us assume that H_0 is true *i.e.*, there is no significant difference between μ_1 and μ_2

Since the sample sizes are large, the test statistic is

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Here $n_1 = 40, n_2 = 30, \bar{x}_1 = 247.3, \bar{x}_2 = 254.1, \sigma_1 = s_1 = 15.2$ and $\sigma_2 = s_2 = 12$.

$$\begin{aligned} \therefore z &= \frac{247.3 - 254.1}{\sqrt{\frac{(15.2)^2}{40} + \frac{(12)^2}{30}}} = \frac{-6.8}{\sqrt{5.776 + 4.8}} \\ &= \frac{-6.8}{\sqrt{10.576}} = \frac{-6.8}{3.2521} = -2.091 \end{aligned}$$

$$\therefore |z| = 2.091 > 1.96$$

Since the computed value of z is greater than the table value, we reject the Null Hypothesis at 5% level and conclude that the two average cars are significantly different *i.e.*, they are not the same in the two busy intersections.

Example 14 : In a certain factory there are two independent processes for manufacturing the same item. The average weight in a sample of 700 items produced from one process is found to be 250 gms with a standard deviation of 30 gms while the corresponding figures in a sample of 300 items from the other process are 300 and 40. Is there significant difference between the mean at 1% level.

[JNTU (H) Apr. 2012 (Set No.1)]

Solution : Let the average weight in the two independent processes be μ_1 and μ_2 respectively.

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$

Then the Alternative Hypothesis is $H_1 : \mu_1 \neq \mu_2$

Let us assume that H_0 is true

i.e., there is no significant difference between μ_1 and μ_2 .

The test statistic is
$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Here $\bar{x}_1 = 250, \bar{x}_2 = 300, \sigma_1 = 30, \sigma_2 = 40$ and $n_1 = 700, n_2 = 300$

$$\therefore z = \frac{250 - 300}{\sqrt{\frac{900}{700} + \frac{1600}{300}}} = \frac{-50}{\sqrt{\frac{9}{7} + \frac{16}{93}}} = -19.43$$

Thus $|z| = 19.43 > 2.58$

Since the computed value of z is greater than the table value, we reject the Null Hypothesis at 1% level and conclude that there is a significant difference between the means.

Example 15 : The mean height of 50 male students who participated in sports is 68.2 inches with a S.D of 2.5. The mean height of 50 male students who have not participated in sport is 67.2 inches with a S.D of 2.8. Test the hypothesis that the height of students who participated in sports is more than the students who have not participated in sports.

[JNTU (H) Apr. 2012 (Set No. 2)]

Solution : Let the mean height in the two cases be μ_1 and μ_2 respectively.

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$

Then the Alternative Hypothesis is $H_1 : \mu_1 \neq \mu_2$

Let us assume that H_0 is true, i.e., there is no significant difference between μ_1 and μ_2

The test statistic is
$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Here $\bar{x}_1 = 68.2, \bar{x}_2 = 67.2, \sigma_1 = 2.5, \sigma_2 = 2.8, n_1 = 50, n_2 = 50$

$$\therefore z = \frac{68.2 - 67.2}{\sqrt{\frac{(2.5)^2}{50} + \frac{(2.8)^2}{50}}} = \frac{1}{\sqrt{\frac{6.5}{50} + \frac{7.84}{50}}} = 1.88$$

$$\therefore |z| = 1.88 < 1.96$$

Since the computed value of z is less than the tabulated value of z , we accept the Null Hypothesis at 5% level of significance and conclude that there is no significant difference in the heights.

Example 16 : The nicotent in milligrams of two samples of tobacco were found to be as follows. Find the standard error and confidential limits for the difference between the means at 0.05 level.

Sample A	24	27	26	23	25	
Sample B	29	30	30	31	24	36

[JNTU (H) Apr. 2012 (Set No. 4)]

Solution :

Calculations for means and s_1^2, s_2^2					
Sample A			Sample B		
x_1	$x_1 - \bar{x}_1$ $= x_1 - 25$	$(x_1 - \bar{x}_1)^2$	x_2	$x_2 - \bar{x}_2$ $= x_2 - 30$	$(x_2 - \bar{x}_2)^2$
24	-1	1	29	-1	1
27	2	4	30	0	0
26	1	1	30	0	0
23	-2	4	31	1	1
25	0	0	24	-6	36
			36	6	36
$125 = \sum x_1$		10	$180 = \sum x_2$		74

$$\bar{x}_1 = \frac{\sum x_1}{n_1} = \frac{125}{5} = 25, \quad \bar{x}_2 = \frac{\sum x_2}{n_2} = \frac{180}{6} = 30$$

$$\text{Hence variance, } s_1^2 = \frac{\sum (x_1 - \bar{x}_1)^2}{n_1 - 1} = \frac{10}{4} = 2.5$$

$$\text{and variance, } s_2^2 = \frac{\sum (x_2 - \bar{x}_2)^2}{n_2 - 1} = \frac{74}{5} = 14.8$$

$$\therefore \text{S. E. of } (\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \sqrt{\frac{2.5}{5} + \frac{14.8}{6}} = 1.72$$

Hence the 95% confidence limits are

$$(\bar{x}_1 - \bar{x}_2) \pm 1.96 (\text{S. E. of } \bar{x}_1 - \bar{x}_2)$$

$$\text{i.e., } (25 - 30) \pm 1.96 (1.72) \text{ i.e., } -5 \pm 3.37 \text{ or } (-8.37, -1.63)$$

REVIEW QUESTIONS

1. Define population and sample.
2. Define small and large samples. [JNTU (A) Dec. 2018]
3. What is testing? What is the purpose of testing? [JNTU (A) June 2018]
3. (a) Define statistical Hypothesis [JNTU (A) Dec. 2018]
4. (b) Define (i) Test of statistical hypothesis.
(ii) Type-I and Type II error.
5. (a) Write about (i) Null hypothesis (ii) Type I and Type II errors (iii) Alternative hypothesis. [JNTU 2005, 2008S (Set No. 1)]
(OR) Explain briefly the following :
(i) Type I error (ii) Type II error (iii) Critical Region
(b) Explain the types of errors in sampling.
(c) Define Null hypothesis and Alternative hypothesis. [JNTU (A) June 2016]
6. What is meant by level of significance. [JNTU (H) Nov. 2009 (Set No. 3)]
(or) Explain the level of significance. [JNTU (A) Dec. 2018]
7. Explain Alternative Hypothesis. [JNTU (A) Dec. 2015]
8. What is the formula of confidence interval? [JNTU (A) Dec. 2015]
9. (a) Write about (i) Critical region (ii) Left tailed test (iii) Right tailed test
(iv) Two tailed test. (or) Write about one tailed and two tailed tests. [JNTU 2004, (H) May 2012 (Set No. 1)]
(b) What is meant by Level of significance, one tailed and two tailed tests? [JNTU (K) Nov. 2011 (Set No. 4)]
(c) Write about (i) Null hypothesis (ii) Critical region (iii) Level of significance.
(d) Explain the critical region and accepting region [JNTU (A) June 2018]
10. Write about null hypothesis and testing of null hypothesis. [JNTU 2005 (Set No. 4)]
11. Explain the procedure generally followed in testing of hypothesis. [JNTU (H) Nov. 2009 (Set No. 2,4), (K) Nov. 2011]
(OR) Explain working rule for testing of Hypothesis. [JNTU (K) Nov. 2011 (Set No. 1)]
12. Write the test statistic in case of single mean for large samples. [JNTU (A) Dec. 2017]
13. Write short notes on type I and type II Error. [JNTU (H) Nov. 2009 (Set No. 2)]
(OR) Explain the types of errors in sampling [JNTU (A) Nov. 2010 (Set No. 4)]
(OR) Explain type I and type II of errors in testing of hypothesis
14. Differentiate two-tailed test of hypothesis from one-tailed test.